

PRIMARY TREATMENT

Technologies of the First Treatment Stage

Wastewater and Solid Waste Management

Institute of Leather Engineering and Technology · University of Dhaka

● Why a Footwear Engineer Needs This

Wastewater treatment is not just a tannery topic — it reaches footwear engineering in several direct ways.

- **Your own factory produces effluent** — finishing, washing, adhesive/cleaning and utility water all need treatment before discharge.
- **You select and audit suppliers** — leather and textile suppliers must run effluent plants to earn LWG / ZDHC status; you must understand what you are auditing.
- **Brands and buyers demand compliance** — ZDHC Wastewater Guidelines, ISO 14001 and buyer codes apply across the footwear supply chain.
- **The law sets discharge limits** — any factory discharging wastewater must meet national standards (Bangladesh ECR 2023).
- **It saves money** — treating and reusing water, and reducing waste, cuts cost and risk.

In short: understanding treatment makes you a better factory engineer AND a credible supply-chain and compliance professional.

● Where Footwear-Factory Wastewater Comes From

Footwear making is mostly a dry, mechanical process — but several wet operations generate effluent.

- **Finishing** — edge inking, dyeing, spraying and cleaning of finishing lines.
- **Component washing & dyeing** — washing uppers/parts; in-house textile or lace dyeing.
- **Adhesive & solvent areas** — cleanup of water-based adhesives, primers and equipment.
- **Rubber / sole production** — compounding and mould/equipment cleaning water.
- **Utilities & general** — boiler blowdown, cooling water, floor washdown and sanitary wastewater.

Note: volumes are far smaller than a tannery's, but oil, grease, finishing chemicals and solids still must be removed.

● What Is in the Wastewater?

Typical footwear-factory effluent — and the general discharge limit it must meet.

Pollutant	Where it comes from	ECR 2023 limit
Suspended solids (SS)	Dust, fibres, finishing residues	150 mg/L
Oil & grease	Adhesives, oils, fats, solvents	10 mg/L
BOD / COD	Adhesives, dyes, organics	50 / 200 mg/L
Colour	Dyes, edge inks, finishes	150 Hazen
pH (variable)	Cleaning & finishing chemicals	6 – 9

Key difference from tanneries: little/no chromium or sulphide, but OIL & GREASE and finishing chemicals are the signature footwear pollutants.

● The Treatment Train — Where Primary Sits

Wastewater is cleaned in stages; each removes what the previous could not.

- **Primary (this lecture)** — physical & physico-chemical: removes solids, oil & grease, balances flow, corrects pH.
- **Secondary** — biological: micro-organisms remove the dissolved biodegradable organics (BOD).
- **Tertiary** — polishing: colour, residual COD, disinfection, and water for reuse.

Role of primary: it does not fully clean the water — it removes the easy, bulky pollutants and PROTECTS the later stages.

● Objectives of Primary Treatment

Four aims — each met by a specific technology.

- **1. Remove coarse & floating matter** — screens and traps take out debris, fibres, oil and grease.
- **2. Balance flow and load** — an equalisation tank smooths the batchy footwear discharge into a steady feed.
- **3. Correct the pH** — neutralise acidic/alkaline cleaning and finishing waters.
- **4. Remove suspended solids** — settling and flotation take out the bulk of the solids and part of the COD.

Together: primary treatment turns a variable, dirty, oily stream into a uniform, screened, pH-corrected feed.

● Primary Treatment Technologies — the Toolbox

The universal unit operations, in the usual order of use.

1

Screening

remove debris & fibres

2

Grit removal

settle out sand/grit

3

Oil & grease removal

skimming / flotation (DAF)

4

Equalisation

balance flow, load & pH

5

Neutralisation

correct the pH

6

Coagulation–flocculation

clump fine solids & colour

7

Sedimentation

settle solids in a clarifier

8

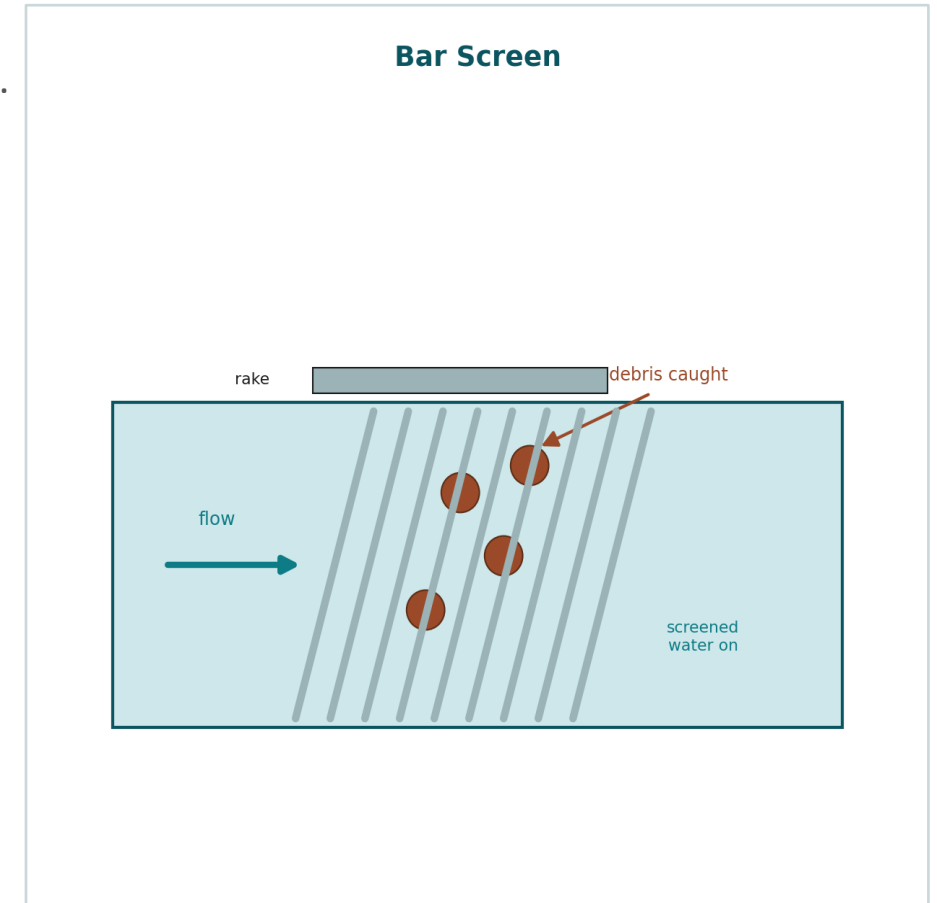
Sludge handling

thicken & dewater solids

● 1 · Screening

The first mechanical step — intercepting coarse and fibrous matter.

- **Coarse (bar) screens** — parallel bars ~20–50 mm apart; cleaned by rakes.
- **Fine screens** — ~1 mm openings; catch fibres, threads and finishing residues.
- **Removes** — thread, fabric scraps, dust clumps, packaging debris.
- **Why** — protects pumps and pipes and takes out 30–40% of the gross solids.



● 2 · Grit Removal

Taking out heavy, gritty solids before they damage the plant.

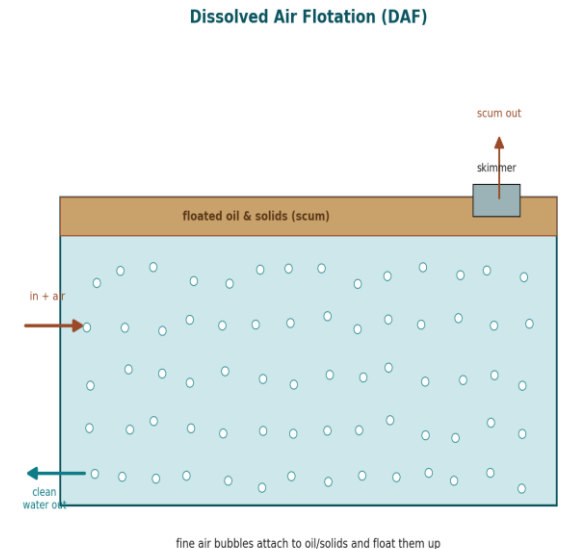
- **What is grit?** — sand, dust and heavy fine particles washed in from floors and components.
- **How** — a grit chamber slows the water so dense grit settles while lighter organics stay suspended.
- **Why it matters** — grit abrades pumps and impellers and fills tanks with dead volume.

Placement: usually right after coarse screening, before pumping and fine screening.

● 3 · Oil & Grease Removal (the footwear priority)

Footwear effluent carries oils, fats, adhesives and solvent residues that must be removed early.

- **Skimming / grease trap** — a quiet zone lets free oil and grease float and be skimmed off the surface.
- **Dissolved Air Flotation (DAF)** — fine air bubbles attach to oil droplets and fine solids and float them to the top as a scum, which is skimmed.
- **Why DAF suits footwear** — it removes emulsified oil and light solids that will NOT settle by gravity.



Important: oil & grease must be removed early — it fouls pipes, coats tanks, and poisons the biological stage.

● 4 · Equalisation

A holding tank that smooths out the ups and downs of factory discharge.

- **The problem** — footwear operations are batchy: finishing, washing and cleanups release sudden slugs of flow, load and pH.
- **What it does** — stores and mixes the wastewater so a steady, averaged stream feeds the rest of the plant.
- **Functions** — flow balancing, load averaging, and time to neutralise the pH.
- **Design rule** — tank volume roughly equals the factory's daily discharge; kept mixed by air or stirrers.

Think of it as: a shock absorber — without it, every batch dump would jolt the downstream units.

● 5 · Neutralisation (pH Correction)

Bringing the mixed effluent to a safe, near-neutral pH.

- **Why** — cleaning agents and finishing chemicals make the water too acidic or too alkaline; biology needs pH 6.5–8.5 and the discharge limit is 6–9.
- **To raise pH** — dose lime or caustic (alkali).
- **To lower pH** — dose a mineral acid (or carbon dioxide).
- **Control** — a pH meter automatically doses the right reagent to hit the target.

Picture it as: balancing a see-saw — add the opposite reagent until the water reaches the neutral middle.

● 6 · Coagulation–Flocculation

A chemical step that clumps fine, non-settling particles into settleable flocs.

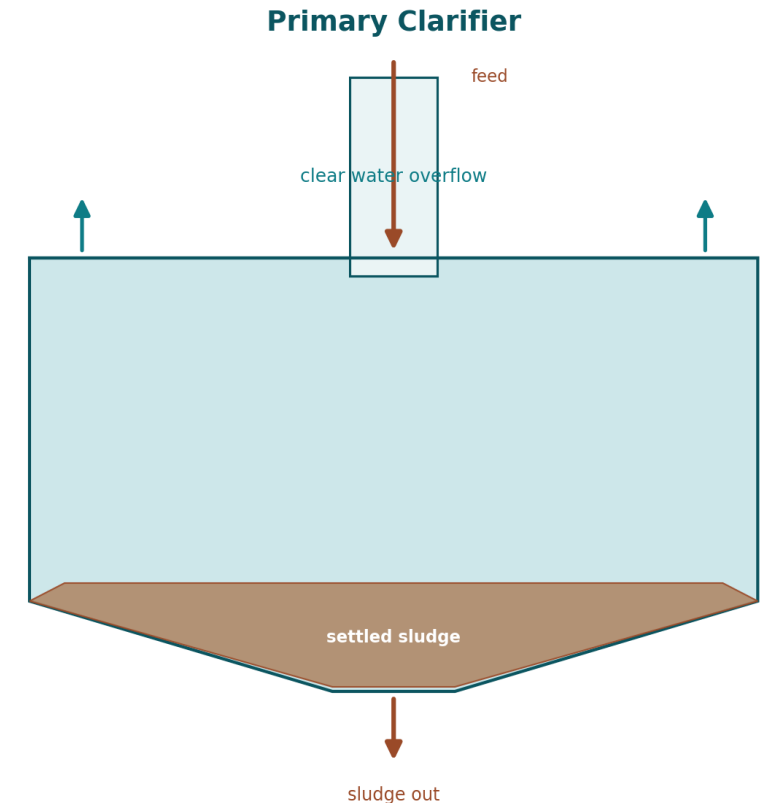
- **The problem** — fine colloids and dye/colour particles are too small and too charged to settle on their own.
- **Coagulation** — a metal salt (alum or ferric chloride) neutralises their charge in a fast mix, so they can come together.
- **Flocculation** — a slow, gentle mix grows them into large flocs that will settle or float.
- **For footwear** — removes colour from dyes/edge-inks, fine solids and part of the COD before settling.

Analogy: like lemon curdling milk — the coagulant makes scattered particles gather into removable lumps.

● 7 · Primary Sedimentation (Clarifier)

Separating the flocs and solids from the water by gravity.

- **How it works** — in a quiet tank, solids sink to the bottom while clear water overflows the edge.
- **Detention time** — about 1–2 hours for the solids to settle.
- **Sludge removal** — a slow scraper pushes settled sludge to a central hopper for withdrawal.
- **Result** — a clarified effluent ready for biological (secondary) treatment.



● 8 · Sludge from Primary Treatment

Cleaning the water concentrates the pollution into sludge, which must be handled.

- **Raw sludge is watery** — only a few percent solids; it must be thickened and dewatered.
- **Dewatering** — by filter press, belt press or drying beds to reduce volume and cost.
- **Oily float (from DAF)** — skimmed oil/grease is collected separately for disposal or recovery.
- **Disposal** — dewatered cake goes to approved landfill, incineration or, where possible, reuse.

Principle: you never destroy pollution in primary treatment — you concentrate it, then manage the concentrate responsibly.

● Primary Treatment at a Glance

Technology	What it removes / does
Screening	Debris, fibres, thread (30–40% of solids)
Grit removal	Sand and heavy grit that abrade pumps
Oil & grease removal / DAF	Floating and emulsified oil, fats, light solids
Equalisation	Balances flow, load and pH (shock absorber)
Neutralisation	Corrects pH to 6–9
Coagulation–flocculation	Clumps fine solids and colour into flocs
Sedimentation (clarifier)	Settles the solids out; clarifies the water
Sludge handling	Thickens and dewateres the removed solids

SECONDARY TREATMENT

Biological Treatment of Tannery Effluent

Wastewater and Solid Waste Management

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PART A

Foundations of Biological Treatment

● Where Secondary Treatment Sits

Treatment is a sequence; secondary is the biological heart of it.

- **Primary (before):** physical/chemical — removed solids, sulphide, corrected pH, cut part of the load.
- **Secondary (here):** biological — micro-organisms remove the DISSOLVED biodegradable organic matter (BOD/COD) that primary cannot.
- **Tertiary (after):** polishing — colour, salts, pathogens, reuse.

Key idea: primary removes what you can filter and settle; secondary removes what is dissolved — and only living microbes can do that.

● What Is Secondary (Biological) Treatment?

A definition to build everything on.

- **Definition:** the use of micro-organisms to consume the dissolved and colloidal biodegradable organic matter, converting it into biomass, CO₂, water and (anaerobically) biogas.
- **Goal:** reduce BOD/COD to within discharge limits (BOD to ≤ 50 mg/L) by growing and then removing microbial biomass.
- **The trick:** feed the microbes the pollution, let them multiply, then settle the microbes out as sludge — the pollution leaves with the sludge.
- **Requirement:** the load must be biodegradable and non-toxic — which is exactly what primary treatment ensured.

● The Microbes — the Living Treatment Plant

Secondary treatment is really microbial farming.

- **Bacteria** — the workhorses; oxidise organic matter for energy and growth.
- **Protozoa & rotifers** — graze on dispersed bacteria, polishing the effluent and improving clarity.
- **Fungi & algae** — present in some systems (fungi at low pH/low N; algae in ponds, giving oxygen).
- **Self-regulating community** — a healthy, diverse biomass is the sign of a well-run plant.
- **General reaction:** organic matter + O₂ + microbes → new microbes + CO₂ + H₂O + energy.

● Aerobic vs Anaerobic Treatment

The first great divide in biological treatment — whether oxygen is present.

Feature	Aerobic (with O ₂)	Anaerobic (no O ₂)
End products	CO ₂ + H ₂ O + much new biomass	Biogas (CH ₄ + CO ₂) + little biomass
Best for	Medium-strength effluent	High-strength effluent (high BOD/COD)
Energy	Consumes energy (aeration)	Produces energy (methane)
Sludge produced	High	Low
Speed	Fast	Slower (needs longer time)
Odour / issues	Low odour	H ₂ S odour; sensitive to toxins
Example units	Activated sludge, trickling filter	UASB, anaerobic digester

● Suspended Growth vs Attached Growth

The second great divide — where the microbes live.

- **Suspended growth** — microbes float freely as flocs in the mixed liquor; water and biomass are mixed, then separated by settling. (Activated sludge & variants.)
- **Attached growth (biofilm)** — microbes grow as a slimy film fixed on a solid surface; water flows past and the film eats the pollution. (Trickling filter, RBC, MBBR.)
- **Hybrid systems** — combine both (IFAS, MBBR-in-activated-sludge) for more capacity in the same tank.
- **Why it matters** — suspended systems are compact and flexible; attached systems are robust, tolerate shocks, need less operator skill.

Remember: every biological process is a combination of two choices — aerobic/anaerobic × suspended/attached.

● Microbial Growth & Substrate Use

How microbes turn pollution into more microbes.

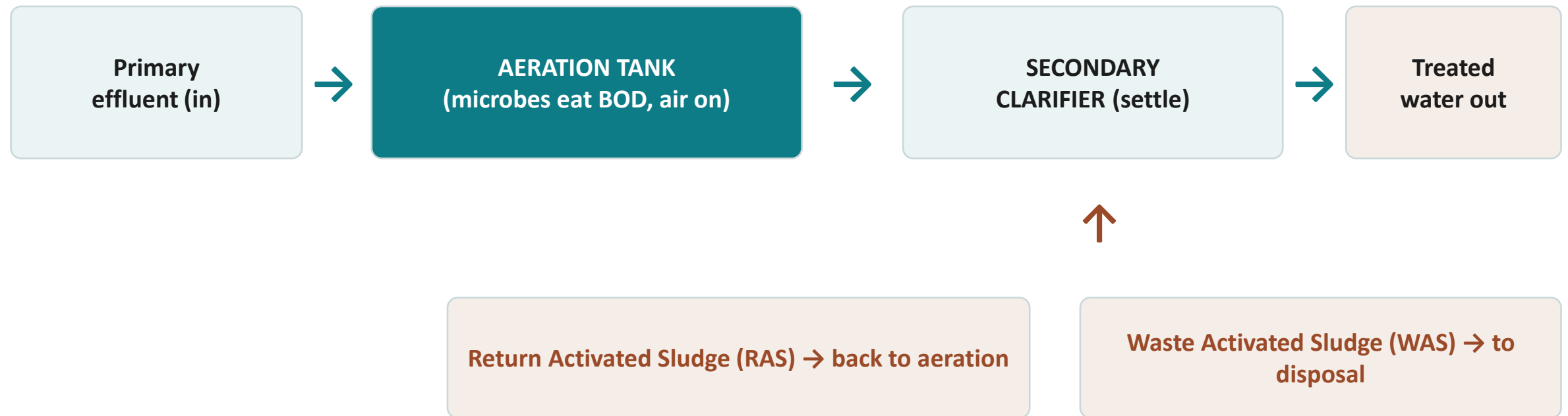
- **Substrate = food** — the organic pollution (BOD) is the microbes' food.
- **Growth** — as they eat substrate, microbes multiply; biomass rises while BOD falls.
- **Endogenous respiration** — when food runs low, microbes consume their own cell mass, reducing and stabilising the sludge.
- **Yield (Y)** — new cells formed per unit substrate consumed; lower yield = less sludge to handle.
- **The balance we control** — feed rate vs biomass (the F/M ratio) sets whether microbes are well-fed or starving.

PART B

Aerobic Suspended Growth — Activated Sludge

● ASP — How It Flows

Follow the water across the top and the sludge loop below.



The loop: keep enough hungry microbes circulating (RAS) and bleed off the surplus (WAS) — that balance runs the whole process.

● The Activated Sludge Process (ASP)

The most widely used secondary treatment in the world.

- **Concept** — wastewater is mixed with a dense population of microbes (activated sludge) in an aerated tank; they eat the BOD and grow into flocs.
- **Two tanks** — an aeration tank (reactor, where eating happens) then a secondary clarifier (where biomass settles out).
- **Recycle (RAS)** — settled biomass is pumped back to the aeration tank to keep the population high.
- **Waste (WAS)** — surplus biomass is removed to keep the population steady.
- **Why 'activated'** — the returned sludge is biologically active and hungry, ready to attack fresh BOD.

● The Control Panel — Five Parameters

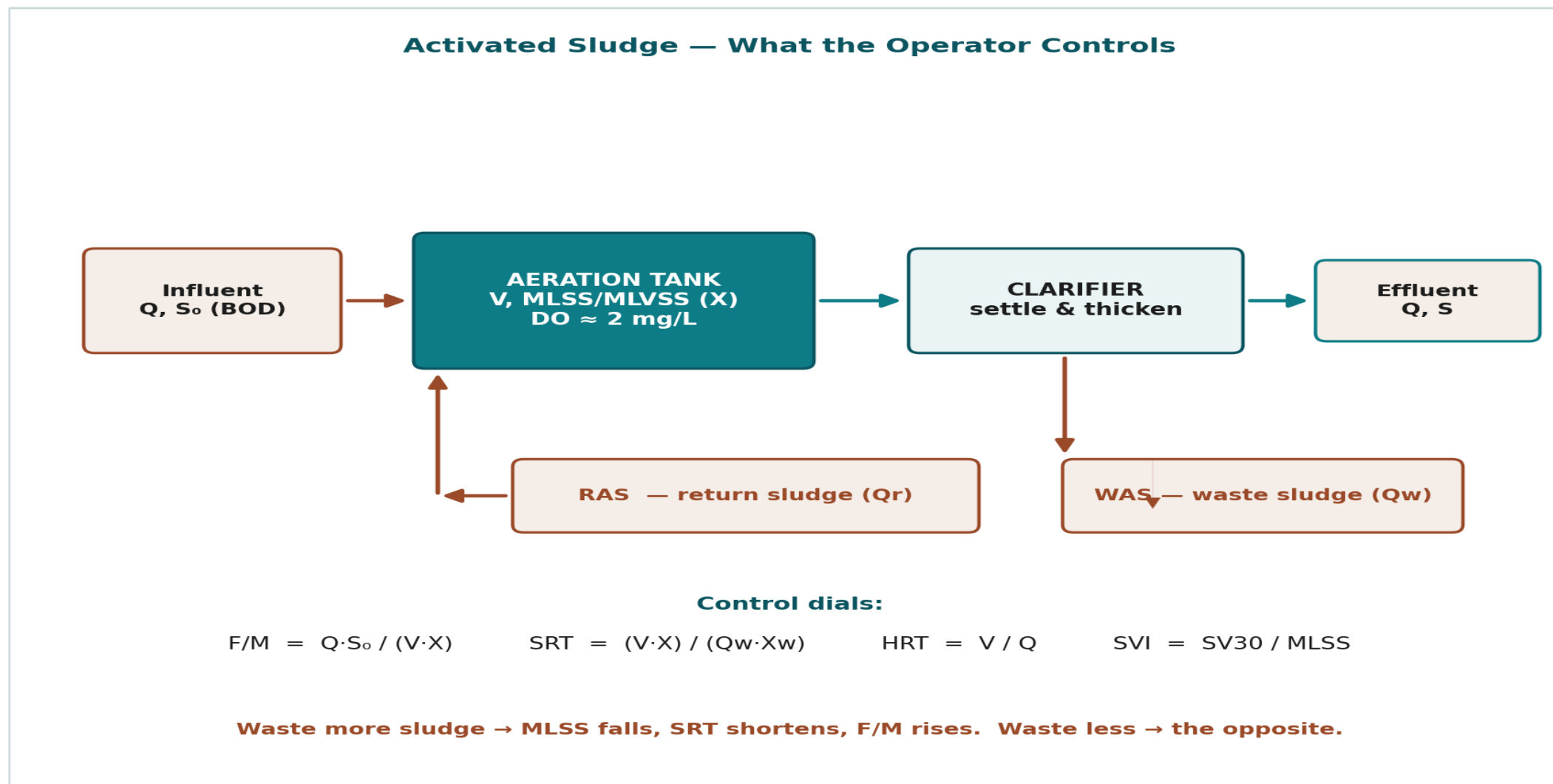
Everything an operator does comes down to these five numbers, plus oxygen.

Parameter	What it means	Typical range	Set by
F/M ratio (Food to Microorganism ratio)	Food per microbe per day	0.2 – 0.5 kg BOD/kg MLVSS·d	MLSS & tank volume
MLSS / MLVSS (Mixed Liquor Suspended Solids/Mixed Liquor Volatile Suspended Solids)	Size of the microbe population	1,500 – 4,000 mg/L	Sludge wasting rate
SRT (Solids Retention Time; sludge age)	How long microbes stay	5 – 15 d (10–30 to nitrify)	Sludge wasting rate
HRT (Hydraulic Retention Time)	How long the water stays	4 – 8 h	Tank volume & flow
SVI (Sludge Volume Index)	How well the sludge settles	< 100 mL/g = good	Diagnosed, not set
DO	Oxygen available to microbes	≈ 2 mg/L	Blower / aerator output

Notice: the sludge-wasting rate (WAS) alone controls BOTH MLSS and SRT — it is the single most powerful lever on the plant.

● Where Each Parameter Lives in the Plant

The flow diagram from Part 1 — now labelled with the quantities we control.



PART A

The Design & Operating Parameters

F/M · MLSS & MLVSS · SRT · HRT · SVI

● Parameter 1 — The F/M Ratio

The food-to-microorganism ratio: how much food each unit of biomass receives per day.

$$F/M = (Q \times S_o) / (V \times X) = \text{kg BOD applied per day} \div \text{kg MLVSS in the tank}$$

- **Q = flow (m³/d)**, **S_o = influent BOD (mg/L)** — together, the FOOD supplied each day.
- **V = aeration tank volume (m³)**, **X = MLVSS (mg/L)** — together, the MICROBES present.
- **Units** — kg BOD / kg MLVSS · day.
- **Design ranges** — conventional 0.2–0.5; extended aeration 0.05–0.15; high-rate 0.4–1.5.

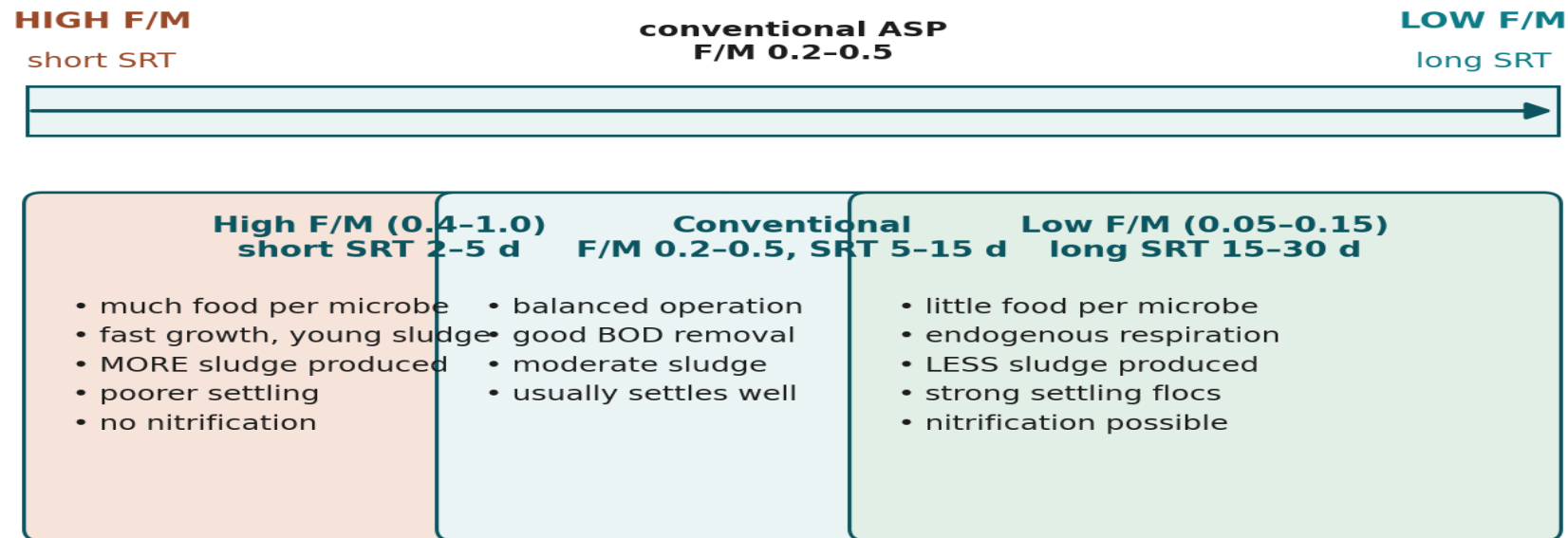
In words: F/M asks — is this herd of microbes underfed, well fed, or overwhelmed by the food arriving today?

Worked example: Q = 1,000 m³/d, S_o = 400 mg/L, V = 500 m³, MLVSS = 2,400 mg/L → F = 400 kg/d, M = 1,200 kg → F/M = 0.33 (normal)

● What F/M Actually Changes

F/M and sludge age are two ends of the same dial — and they set the character of the sludge.

F/M Ratio and Sludge Age (SRT) — Opposite Ends of One Dial



● Reading the F/M Ratio

Each region of the dial produces a recognisably different plant.

- **F/M too HIGH (overloaded)** — microbes cannot consume all the BOD; effluent BOD rises; growth becomes dispersed and flocs settle poorly; sludge production is high. Cure: raise MLSS (waste less) or reduce load.
- **F/M in RANGE (0.2–0.5)** — good BOD removal, well-formed flocs, predictable sludge production. The design target for conventional ASP.
- **F/M too LOW (underloaded)** — microbes starve into endogenous respiration; sludge is old and mineralised with little excess; but pin-point floc and filaments can appear and aeration energy is wasted.
- **Deliberate low F/M = extended aeration** — many industrial and small plants CHOOSE $F/M \approx 0.05\text{--}0.15$ for stability and minimal sludge.

Rule of thumb: raise MLSS to lower F/M; lower MLSS to raise it — and you control MLSS by how much sludge you waste.

● Parameter 2 — MLSS and MLVSS

How we measure the size of the microbial population in the aeration tank.

- **Mixed liquor** — the mixture of wastewater and activated sludge inside the aeration tank.
- **MLSS (Mixed Liquor Suspended Solids)** — ALL suspended solids in the mixed liquor: living microbes + dead cells + inert matter. Measured by filtering and drying at 105 °C.
- **MLVSS (Volatile Suspended Solids)** — the fraction that burns off at 550 °C, i.e. the ORGANIC (essentially living) part. Typically 70–80% of MLSS.
- **Why both are needed** — MLVSS is the true 'M' in F/M (only living biomass eats), but MLSS is faster to measure, so plants track MLSS daily and apply the MLVSS/MLSS ratio.
- **MBR plants** — run far higher, 8,000–12,000 mg/L, because a membrane (not gravity) separates the solids.

$$\text{MLVSS} \approx 0.7 - 0.8 \times \text{MLSS}$$

Control: waste more sludge → MLSS falls.
Waste less sludge → MLSS rises.

● Parameter 3 — Sludge Age (SRT)

The average time a microbe spends in the system — the master variable of biological design.

$$\text{SRT} = (V \times X) / (Q_w \times X_w) = \text{kg biomass in system} \div \text{kg biomass wasted per day}$$

- **Also called** — solids retention time, mean cell residence time (MCRT), or simply sludge age.
- **Short SRT (2–5 d)** — young, fast-growing biomass; high sludge yield; carbon removal only.
- **Medium SRT (5–15 d)** — conventional activated sludge; reliable BOD removal.
- **Long SRT (15–30 d)** — extended aeration; stable, minimal excess sludge, well mineralised.
- **SRT selects the species** — any microbe reproducing more slowly than the SRT is washed out. This is WHY sludge age decides what the plant can remove.

Critical rule: nitrifiers grow slowly — nitrification needs $\text{SRT} \gtrsim 10$ days (longer when cold), or ammonia passes straight through.

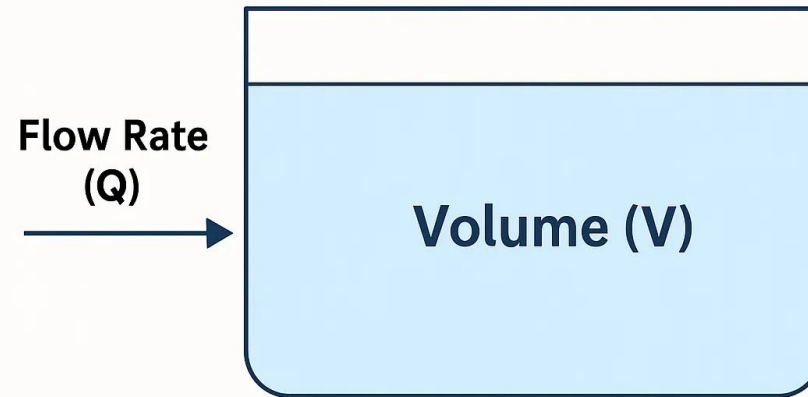
Worked example: biomass in system = 1,200 kg; wasting 100 kg/d → SRT = 12 days (nitrification feasible)

● Parameter 4 — HRT, and Why SRT $\gg$ HRT

How long the WATER stays versus how long the MICROBES stay — the key insight of activated sludge.

Calculating Hydraulic Retention Time

$$\text{HRT} = \frac{V}{Q}$$



The central idea: activated sludge works because the SLUDGE is retained far longer than the WATER. Everything else is refinement.

● Parameter 4 — HRT, and Why SRT $\gg$ HRT

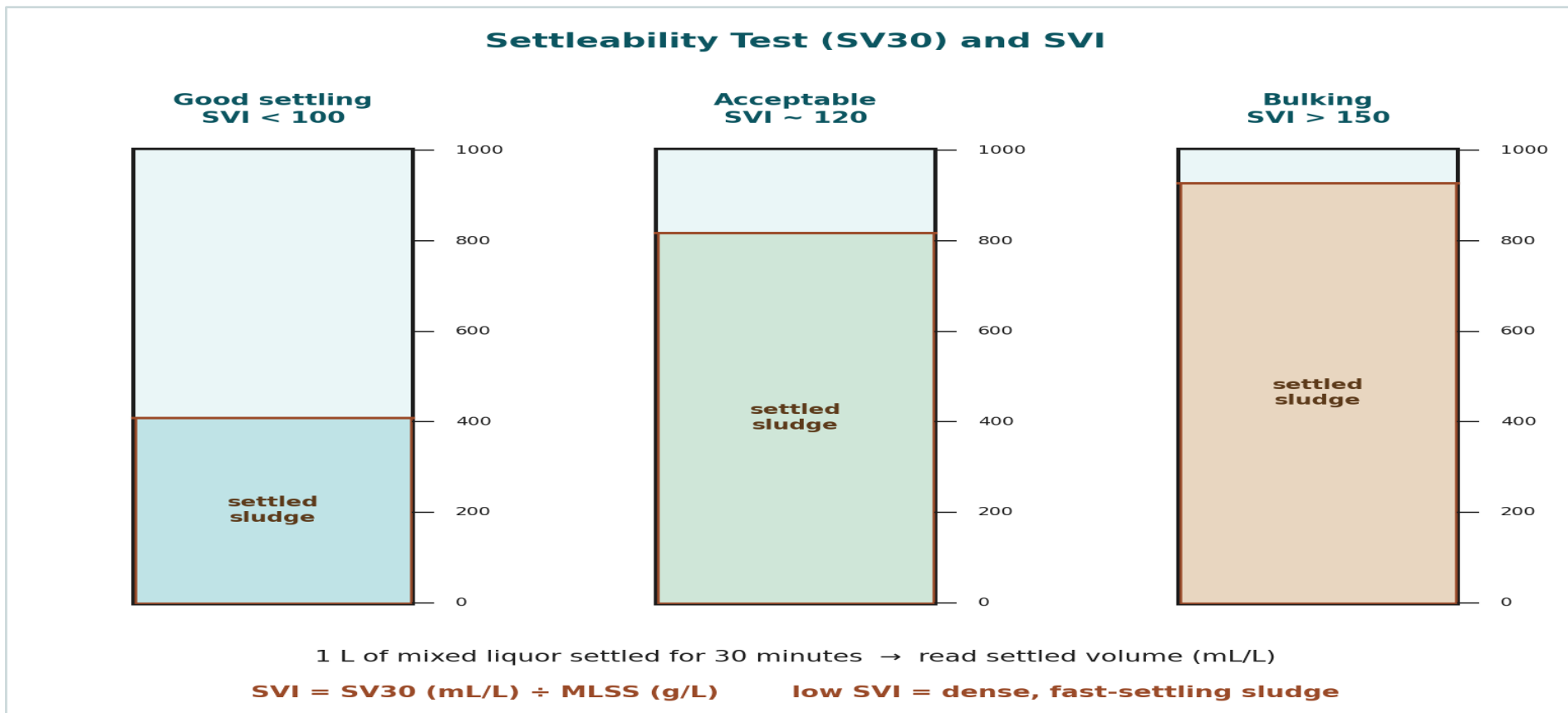
How long the WATER stays versus how long the MICROBES stay — the key insight of activated sludge.

- **HRT (Hydraulic Retention Time)** = V / Q — the time water spends in the aeration tank. Conventional ASP 4–8 hours; extended aeration 18–36 hours.
- **SRT is far longer than HRT** — typically 5–15 DAYS versus 4–8 HOURS, because the clarifier catches the biomass and RAS returns it.
- **Why that matters** — it decouples the two: a large, mature microbial population can be held in a small tank treating a large flow. Without sludge return, slow-growing microbes would simply wash out.
- **Volumetric organic loading** — kg BOD per m^3 per day (0.3–1.0 typical for conventional ASP); the compactness measure of the reactor.

The central idea: activated sludge works because the SLUDGE is retained far longer than the WATER. Everything else is refinement.

● Parameter 5 — Settleability and SVI

Excellent biological treatment is worthless if the biomass will not settle out afterwards.



$$\text{SVI} = \text{SV30 (mL/L)} \times 1000 \div \text{MLSS (mg/L)} \quad \cdot \quad < 100 \text{ good} \cdot 100\text{--}150 \text{ fair} \cdot > 150 \text{ bulking}$$

● The SV30 Test and What SVI Tells You

A one-litre cylinder and thirty minutes — the most informative daily test in the plant.

- **The test (SV30)** — fill a 1 L graduated cylinder with mixed liquor from the aeration tank, settle 30 minutes, read the settled sludge volume in mL/L.
- **Convert to SVI** — divide by MLSS; SVI is the volume in mL occupied by ONE GRAM of dry sludge after 30 minutes of settling.
- **Low SVI (< 100)** — dense, compact, fast-settling floc: the clarifier works and the returned sludge is thick.
- **High SVI (> 150)** — light, bulky sludge that settles slowly and carries over the weir, failing the effluent TSS limit.

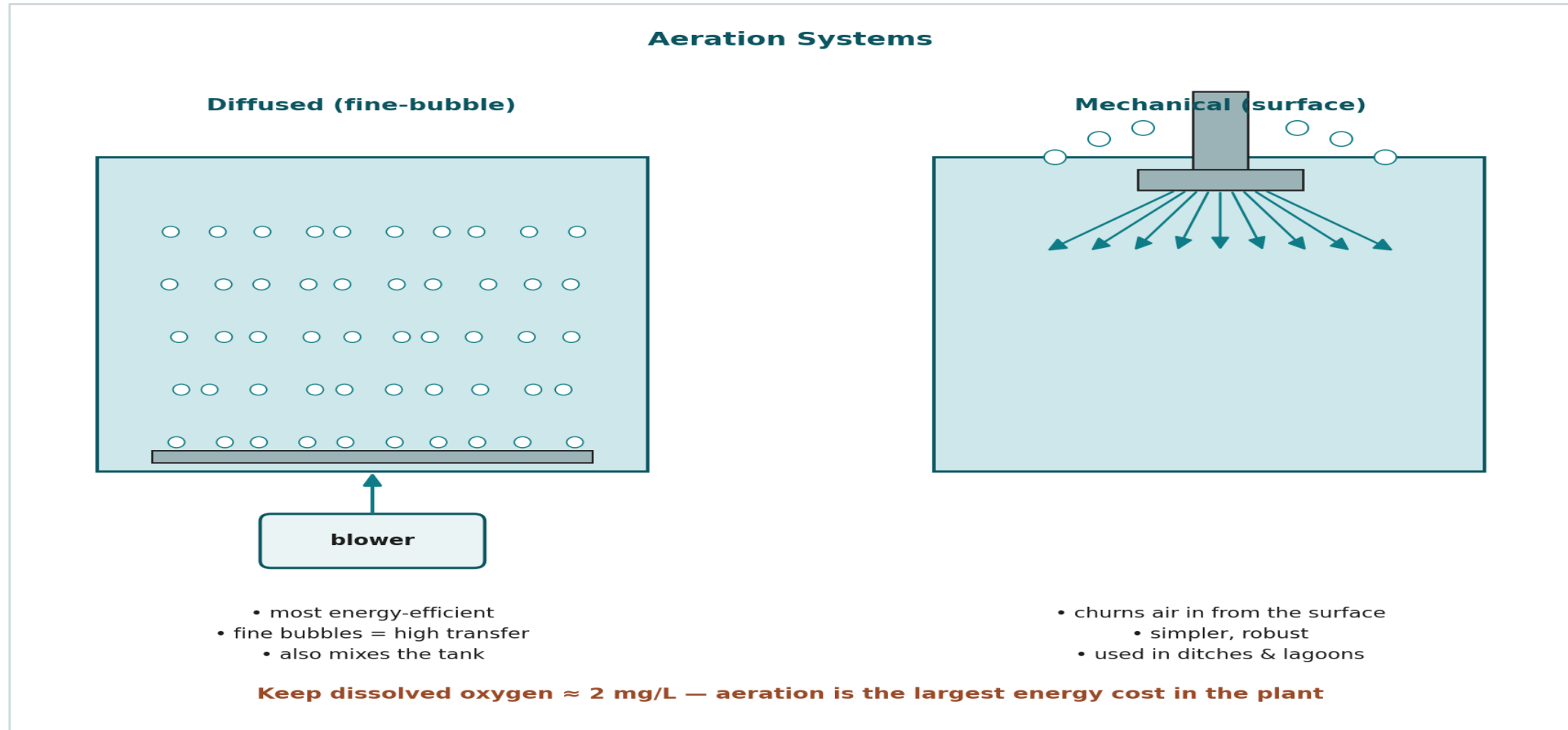
Worked example: $SV30 = 280 \text{ mL/L}$, $MLSS = 2,800 \text{ mg/L}$ → $SVI = 280 \times 1000 \div 2,800 = 100 \text{ mL/g}$ (satisfactory)

Operator's habit: run SV30 daily and examine the sludge under a microscope — the biology warns you before the lab results do.

Also watch: sludge colour (healthy = chocolate brown), odour (earthy, not septic), and floc appearance.

● Oxygen — Aeration Systems

Aerobic microbes must be supplied oxygen continuously; this is the plant's largest energy cost.



● Aeration — Demand, Control and Energy

How much oxygen is needed, how it is delivered, and how the energy bill is managed.

- **Oxygen demand** — roughly 0.8–1.2 kg O₂ per kg BOD removed, plus about 4.6 kg O₂ per kg of ammonia nitrogen oxidised if the plant nitrifies.
- **Target DO ≈ 2 mg/L** — below about 0.5 mg/L treatment suffers and filaments thrive; above about 3 mg/L is wasted energy.
- **Fine-bubble diffused aeration** — highest oxygen-transfer efficiency; membrane diffusers on the tank floor; also provides mixing. Preferred for new plants.
- **Mechanical surface aerators** — simple and robust, used in oxidation ditches and lagoons; lower transfer efficiency, more aerosols and odour.
- **Energy share** — aeration is typically 45–60% of total plant energy, so DO-based blower control repays its cost quickly.

Design note: mixing must also be maintained — the tank needs enough energy input to keep MLSS in suspension even when oxygen demand is low.

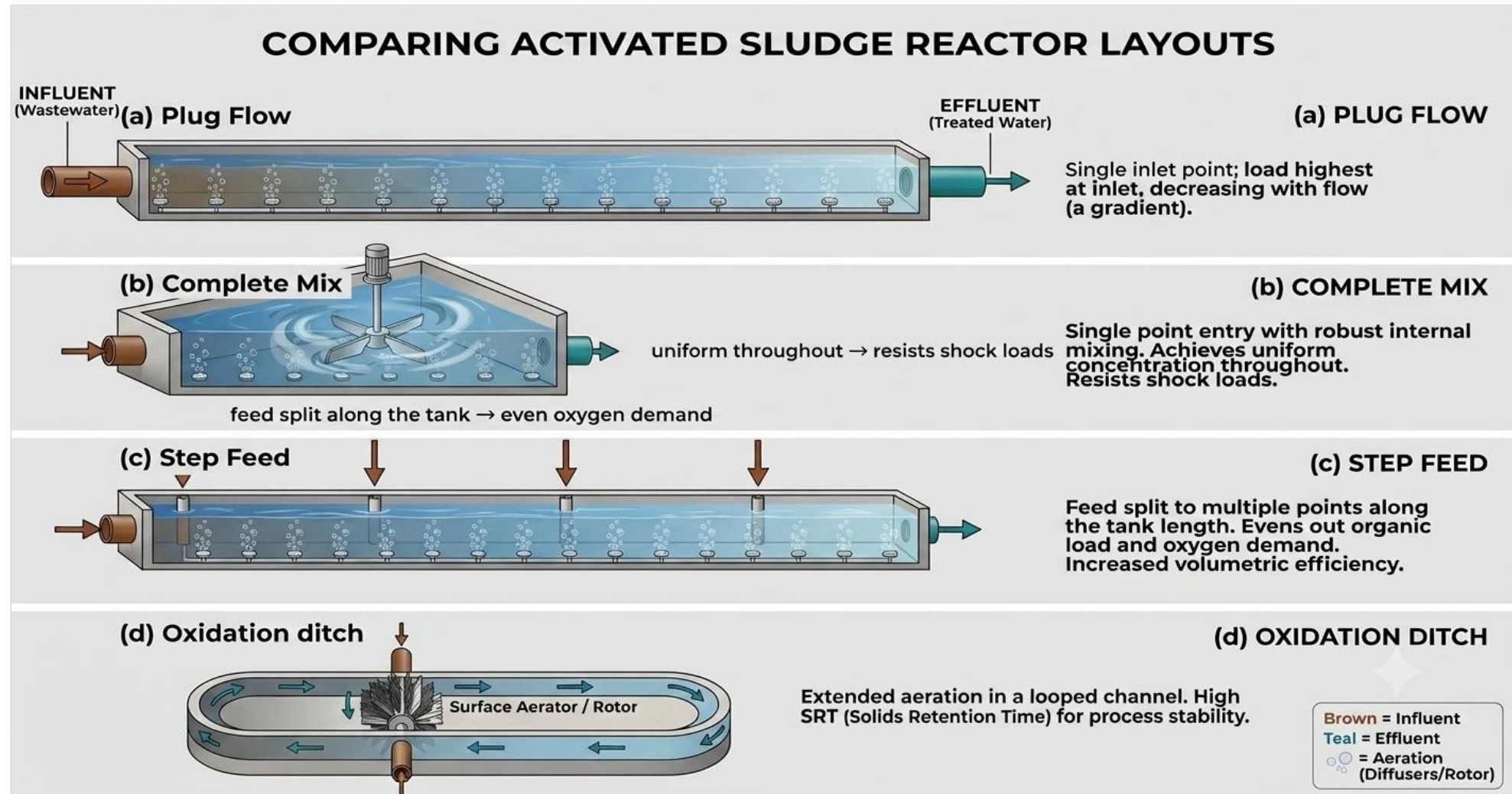
PART B

Process Variants of Activated Sludge

Plug flow · complete mix · extended aeration · step feed · SBR

● Reactor Configurations

The same biology, arranged differently to suit load, land, energy and effluent targets.



● Comparing the Variants

Textbook design ranges — note how F/M and SRT define each one.

Variant	F/M	SRT (d)	HRT (h)	Character
Conventional plug flow	0.2 – 0.4	5 – 15	4 – 8	Standard; load highest at inlet
Complete mix	0.2 – 0.6	5 – 15	3 – 5	Dilutes shocks — good for industry
Extended aeration	0.05 – 0.15	20 – 30	18 – 36	Stable, least sludge, big tank
High-rate	0.4 – 1.5	2 – 5	1.5 – 3	Compact, partial treatment
Step feed	0.2 – 0.4	5 – 15	3 – 5	Feed split — evens oxygen demand
Contact stabilisation	0.2 – 0.6	5 – 15	0.5 – 1 contact	Two tanks; smaller volume
Oxidation ditch	0.05 – 0.15	15 – 30	18 – 30	Looped channel; robust
SBR	0.05 – 0.3	10 – 30	cycle-based	One tank; excellent for N and P

Reading the table: extended aeration and oxidation ditches trade tank volume for stability and low sludge — which is why they suit industrial effluent.

● Extended Aeration — the Industrial Favourite

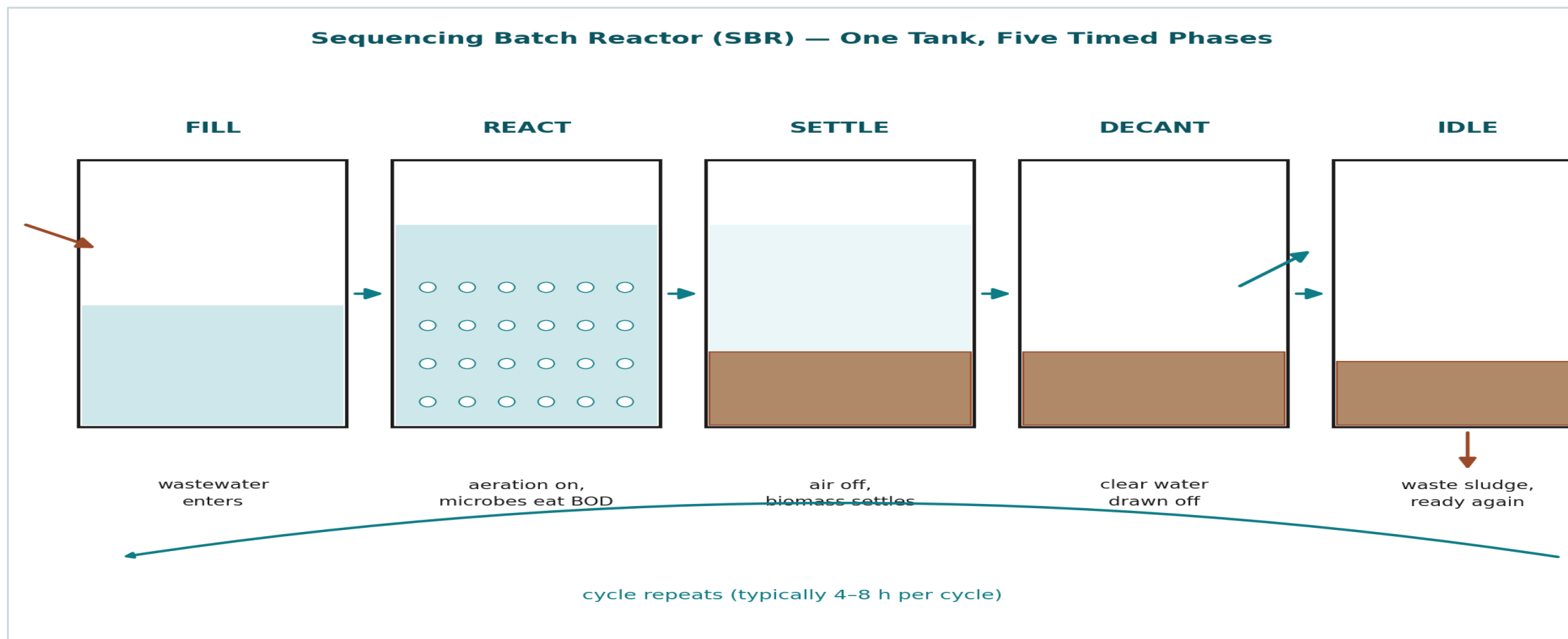
The variant most often chosen for tanneries and small factories — worth understanding in detail.

- **What it is** — activated sludge run deliberately at very low F/M (0.05–0.15) and very long SRT (20–30 d), with HRT of 18–36 hours.
- **Endogenous respiration dominates** — starved microbes consume their own cell mass, so very little excess sludge is produced and what remains is well stabilised (less odour, easier dewatering).
- **Very stable** — the large biomass and long retention absorb shock loads and toxic slugs far better than a high-rate plant.
- **Nitrification comes free** — the long sludge age lets nitrifiers establish, so ammonia is oxidised as well.
- **The price** — a much larger aeration tank and higher aeration energy per unit of BOD removed.

Why tanneries choose it: forgiving, simple to operate, minimal sludge handling — robustness beats efficiency when the influent is variable and hostile.

● The Sequencing Batch Reactor (SBR)

All five steps happen in ONE tank, separated in TIME rather than in space.



The key insight: switching aeration on and off within the React phase creates aerobic, anoxic and anaerobic conditions in one tank — so it can remove C, N and P.

● SBR — Operation, Advantages and Limits

The phases in detail, and why SBRs have become so widely used.

- **Fill (aerated or not)** — influent enters onto the settled sludge; an unaerated fill creates the anaerobic/anoxic conditions needed for phosphorus and nitrogen removal.
- **React** — aeration on; BOD is oxidised and ammonia nitrified. Cycling the air off gives anoxic denitrification within the same phase.
- **Settle** — aeration and mixing stop; the tank becomes a perfectly quiescent clarifier with no inflow disturbance — often settling better than a continuous clarifier.
- **Decant** — a moving weir draws clear supernatant off the top; the sludge stays behind.
- **Idle / waste** — surplus sludge is wasted and the tank is ready for the next cycle (typically 4–8 h).
- **Limitations** — requires reliable automation, level control and decanters; usually two or more tanks alternate so inflow can be continuous.

PART C

Operating Problems & Troubleshooting

Bulking · foaming · rising sludge · toxic shock

● Sludge Bulking — the Classic Failure

The most common and most damaging operating problem in activated sludge.

- **What happens** — filamentous bacteria grow out of the floc, forming a light, open structure that traps water and will not compact. SVI rises above 150 mL/g.
- **Consequence** — the sludge blanket rises in the clarifier, biomass carries over the weir, effluent TSS and BOD fail, and MLSS is progressively lost from the system.
- **Common causes** — low dissolved oxygen; prolonged low F/M (starvation); nutrient deficiency (N or P); septic influent containing sulphide; wide swings in load.
- **Nutrient rule of thumb** — maintain $\text{BOD} : \text{N} : \text{P} \approx 100 : 5 : 1$. Many industrial effluents are nitrogen- or phosphorus-deficient and must be dosed.

Cures, in order: fix DO first, correct nutrients, adjust F/M and SRT, add a selector zone — and only as an emergency measure, chlorinate the return sludge.

Diagnosis: microscopic examination showing filaments bridging between flocs confirms bulking earlier than SVI data alone.

● Other Common Problems — Diagnosis Table

Symptom → likely cause → corrective action. Learn to read the plant.

Symptom	Likely cause	Corrective action
Thick brown foam on tank	Nocardia / actinomycetes; grease; long SRT	Reduce SRT, remove foam physically, control fats & oil
White billowing foam	Very young sludge / low MLSS (start-up)	Reduce wasting, let MLSS build up
Sludge rising in clarifier	Denitrification releasing N ₂ gas in the blanket	Increase sludge withdrawal, reduce clarifier detention
Pin-point floc, turbid effluent	Over-oxidised sludge, F/M too low	Increase F/M (waste more), shorten SRT
Dispersed growth, cloudy effluent	Overloading; F/M too high	Increase MLSS, reduce load
Sudden loss of activity	Toxic shock (chrome, sulphide, solvent slug)	Find & stop source, dilute, reseed biomass
Dark, septic, sulphide odour	Anaerobic conditions, low DO	Increase aeration; check sludge blanket depth

● Toxic Shock — the Industrial Reality

For tannery and industrial effluent, the biggest operating risk is a poisoned biomass.

- **What it looks like** — DO suddenly RISES (the microbes have stopped respiring), effluent BOD climbs, the sludge goes pale and disperses, and the microscope shows dead or immobile protozoa.
- **Common causes in tanneries** — a slug of chrome liquor, a sulphide-bearing lime dump reaching the aeration tank, a solvent or biocide discharge, or a sudden salinity spike.
- **Immediate response** — identify and stop the source; dilute the tank; maintain aeration; STOP wasting sludge to preserve whatever biomass survives.
- **Recovery** — reseed with healthy sludge from another plant and re-acclimatise gradually; full recovery takes days to weeks.
- **Prevention is the real answer** — strong equalisation, stream segregation, sulphide oxidation and chrome recovery upstream, plus influent monitoring.

Remember: the biology is your most expensive and most fragile asset — and protecting it is largely a primary-treatment job.

● Routine Monitoring — What to Measure and How Often

Good operation is a monitoring discipline, not an occasional intervention.

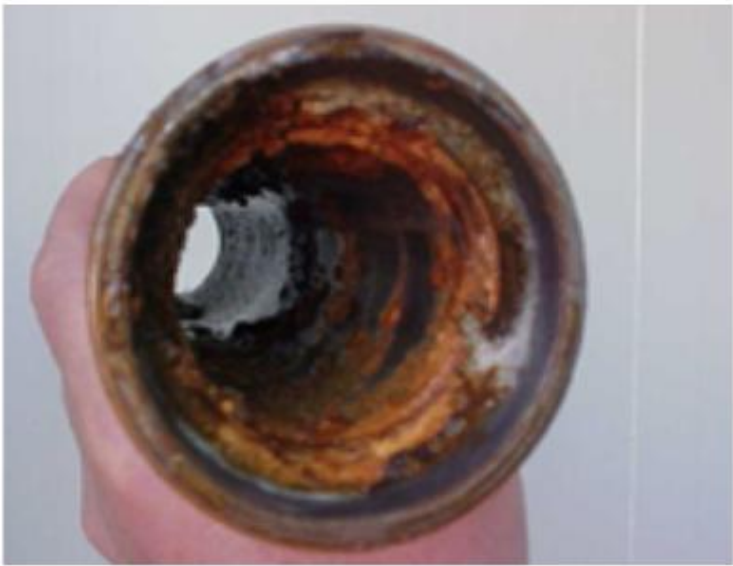
- **Every shift / daily** — DO in the aeration tank, SV30 settling test, sludge appearance and colour, clarifier blanket depth, pH and temperature.
- **Daily to weekly** — MLSS and MLVSS, influent and effluent BOD/COD, effluent TSS; calculate F/M, SRT and SVI; record RAS and WAS flows.
- **Weekly to monthly** — microscopic examination of the sludge, nutrient (N and P) check, ammonia/nitrate if nitrifying, sludge production trend.
- **Track the trend, not just the number** — a single SVI value means little; SVI climbing over five days is an early warning of bulking.

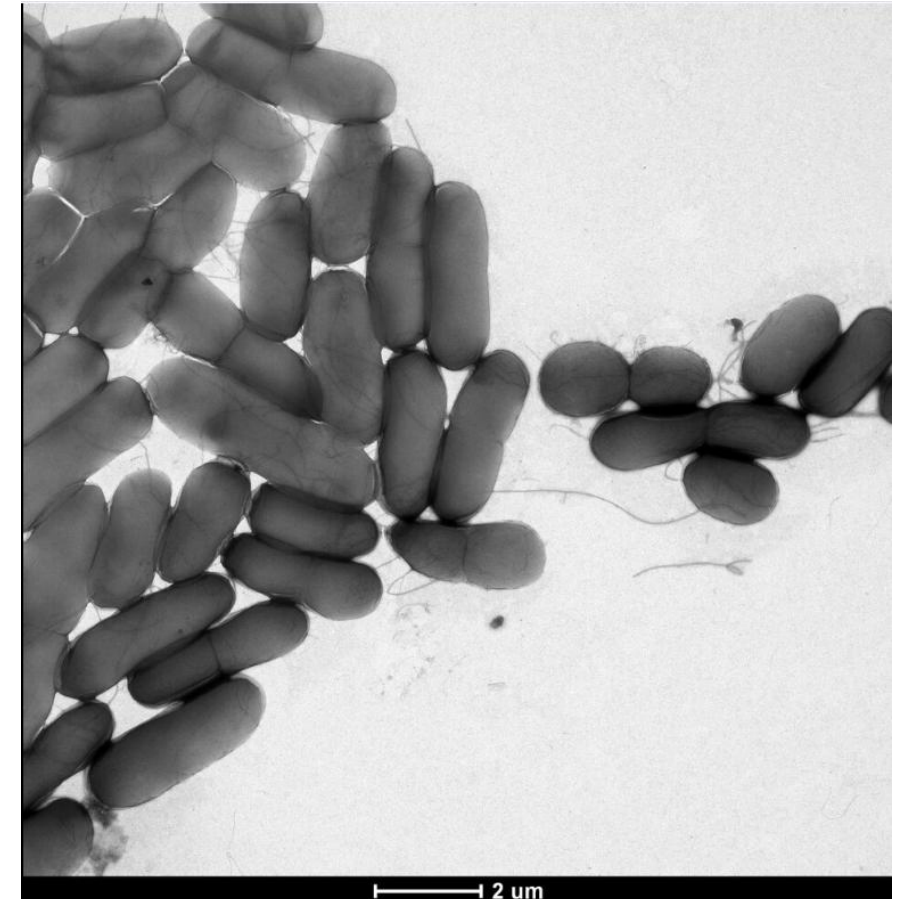
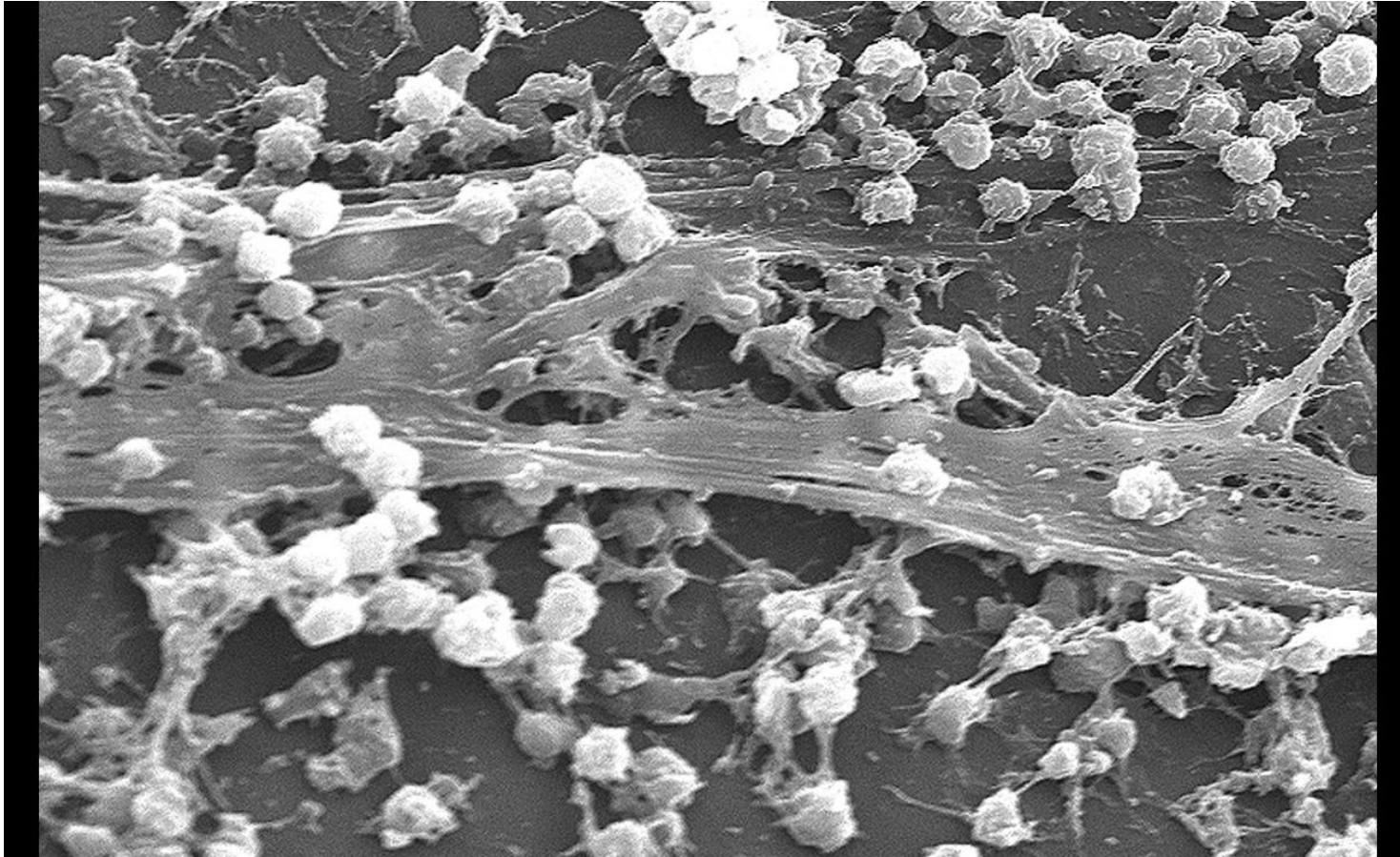
Practical target: plot F/M, SRT, MLSS and SVI on one weekly chart — most failures announce themselves as a trend long before an effluent violation.

PART A

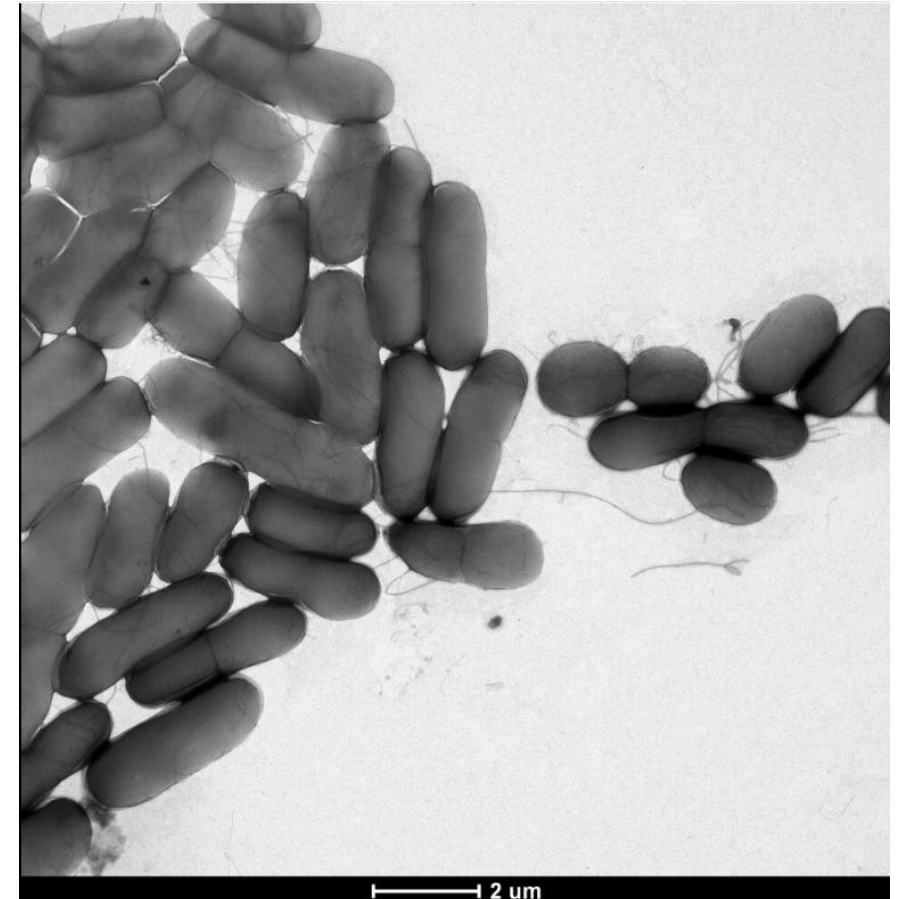
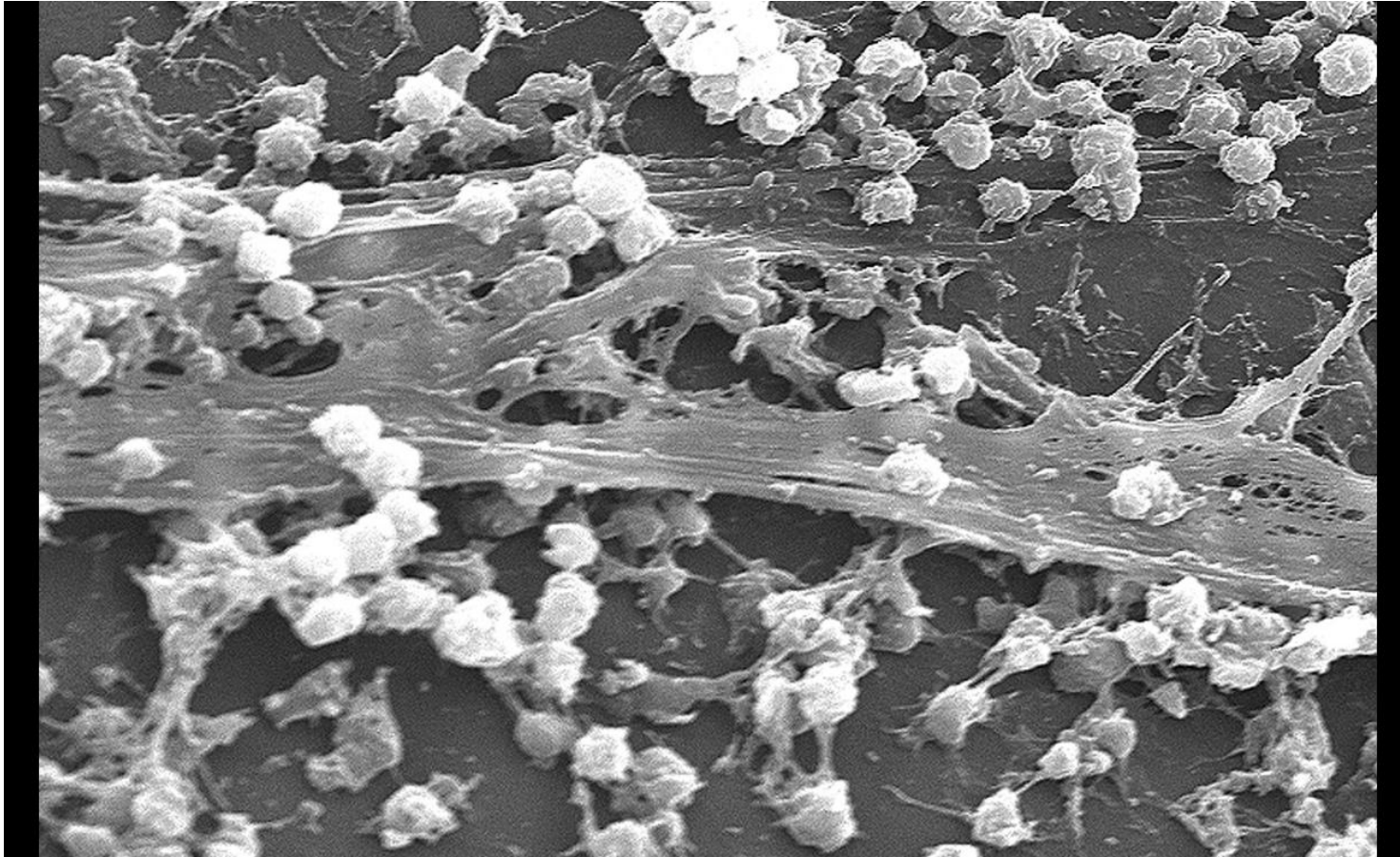
The Biofilm — How Attached Growth Works

Structure · substrate transport · sloughing





● What Is a Biofilm?



● What Is a Biofilm?

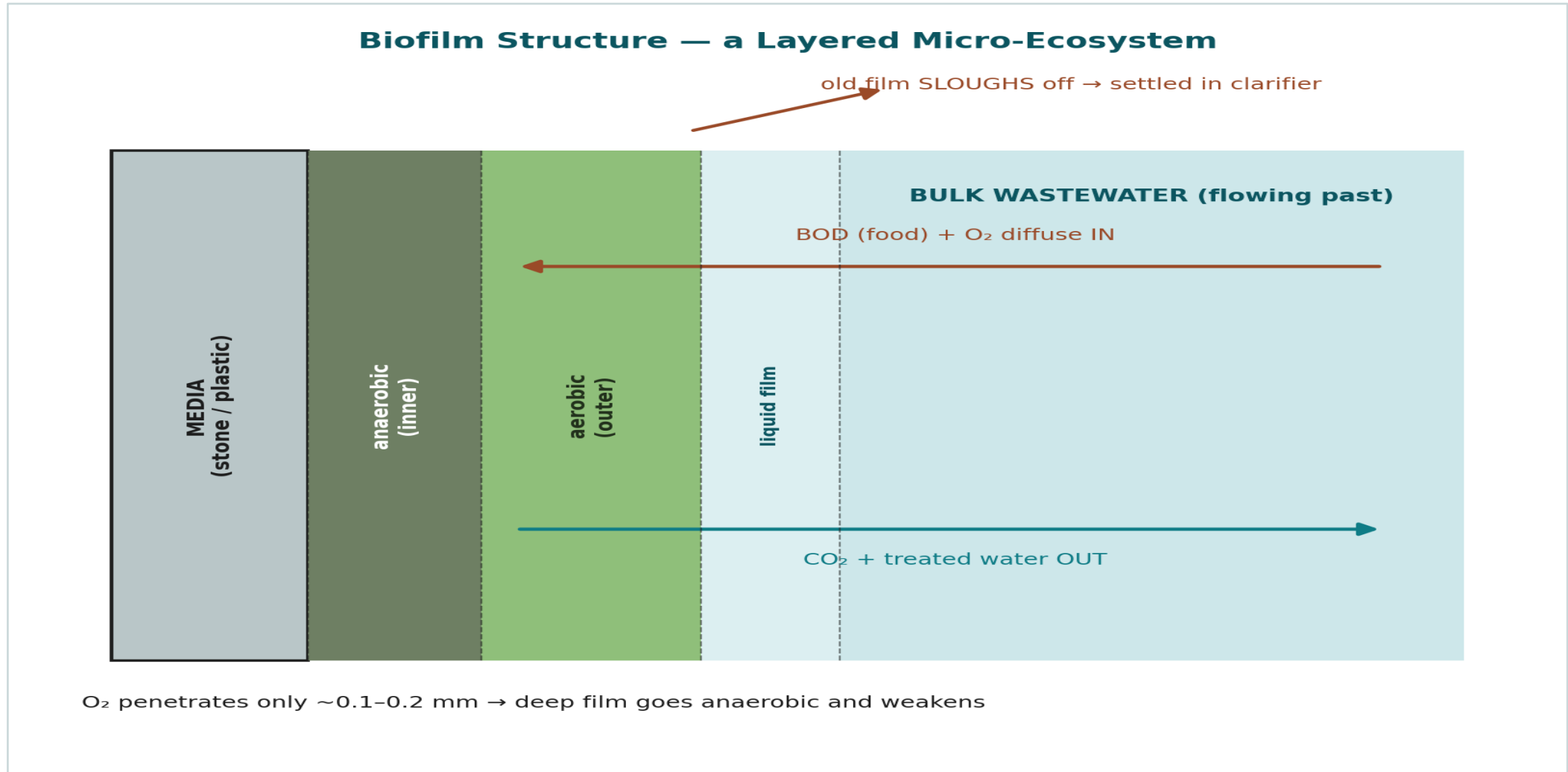
The central object of this lecture — a living film glued to a surface.

- **Definition** — a layer of micro-organisms attached to a solid support and held together in a matrix of extracellular polymeric substances (EPS), the slime the microbes secrete.
- **How it forms** — free bacteria attach to the media, multiply, and excrete EPS which cements them in place; the film thickens over days into a stable community.
- **The community** — bacteria do the BOD removal; protozoa, rotifers, worms and fly larvae graze on them, keeping the film from over-thickening.
- **How it feeds** — it does NOT move to the food; the wastewater flows past and the food (BOD) and oxygen must DIFFUSE into the film.
- **Thickness** — typically 0.1–2 mm; only the outer 0.1–0.2 mm receives oxygen.

Key consequence: because transport is by diffusion, biofilm performance is limited by how deeply food and oxygen can penetrate — not by how thick the film grows.

● Inside the Biofilm — a Layered Structure

Follow the food and oxygen inward, and the products outward.



● Diffusion, Depth and Sloughing

Three mechanisms that govern every attached-growth reactor.

- 1 • Diffusion in** — BOD and dissolved oxygen move from the bulk liquid, across a thin stagnant liquid film, into the biofilm. This transport step is often the rate-limiting one.
 - 2 • Layered metabolism** — the outer film is aerobic and does most of the work; deeper in, oxygen runs out and conditions turn anoxic then anaerobic.
 - 3 • Sloughing** — as the film thickens, the anaerobic base weakens and loses its grip, so patches detach and wash away. This is the biofilm system's 'sludge wasting' — it happens naturally.
- **Why a clarifier is still needed** — sloughed biofilm leaves with the effluent as humus sludge and must be settled out downstream.
 - **Design implication** — thicker film does NOT mean better treatment; what matters is the SURFACE AREA available and the hydraulic contact with it.

Compare with Part 2: in activated sludge YOU choose the sludge age by wasting; in a biofilm the system regulates itself by sloughing.

PART B

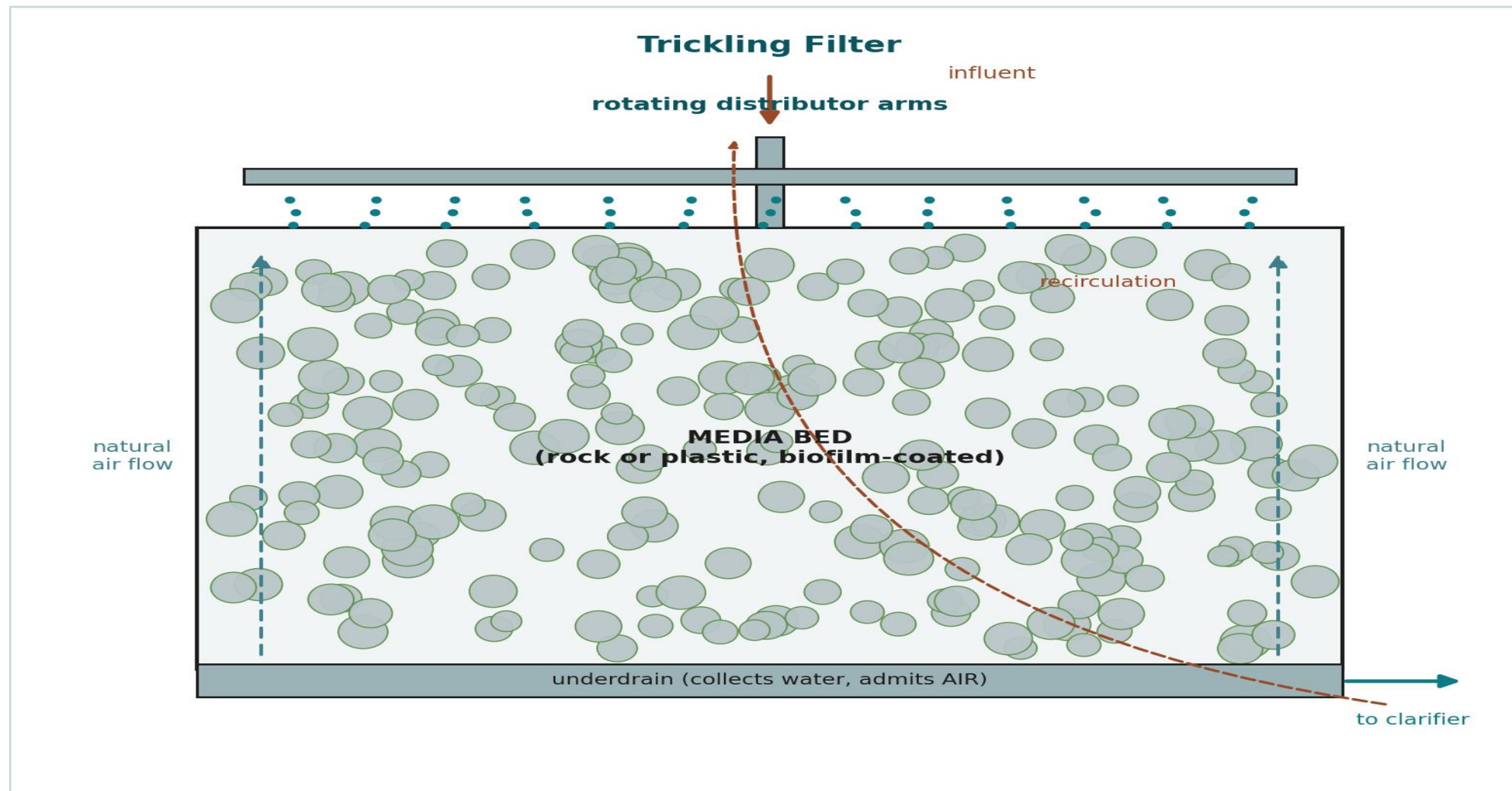
The Trickling Filter

Media · distributor · loading · recirculation



● The Trickling Filter

Despite the name it does not filter — it is a biofilm reactor on a fixed bed of media.



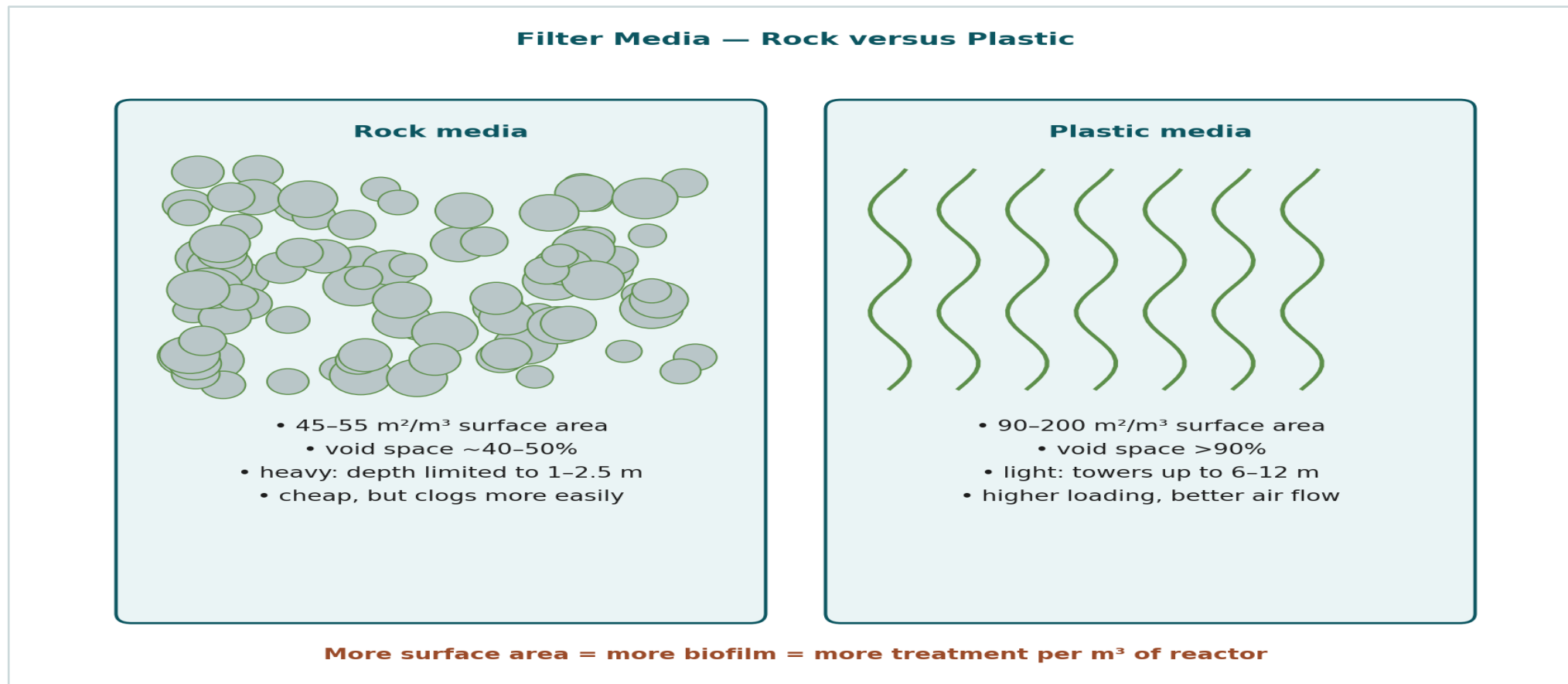
● Trickling Filter — How It Works

The components and the path the water takes.

- **The bed** — a circular tank filled with rock or plastic media, 1–2.5 m deep for rock, up to 6–12 m for plastic towers.
- **The distributor** — rotating arms spray settled wastewater evenly over the surface; the water then trickles down over the biofilm-coated media.
- **Treatment** — as the thin liquid passes over the biofilm, BOD diffuses in and is oxidised; the water emerges progressively cleaner.
- **Ventilation** — air moves naturally through the voids (driven by the temperature difference between air and water), supplying oxygen — no blowers needed for classical designs.
- **Underdrain** — collects the treated water and admits air; it must never run full or ventilation stops.
- **Then a clarifier** — the effluent carries sloughed biofilm (humus sludge), which is settled in a secondary clarifier.

● The Media — Where the Surface Area Comes From

Media choice sets the treatment capacity of the filter.



Why plastic changed the game: 3–4× the surface area and far greater void space allow tall towers, high loading and reliable air flow without clogging.

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● Trickling Filter — Design Parameters

The two loadings that define the design, plus recirculation.

$$\text{Hydraulic loading} = Q / A \text{ (m}^3\text{/m}^2\cdot\text{d)}$$

$$\text{Organic loading} = Q \cdot S_o / V \text{ (kg BOD/m}^3\cdot\text{d)}$$

- **Hydraulic loading** — flow per unit of filter surface area; controls wetting, flushing of excess film and contact time.
- **Organic loading** — BOD applied per unit of media volume per day; controls how much food the biofilm must handle.
- **Recirculation ratio ($R = Q_r/Q$)** — typically 0.5–3.0. Returning effluent dilutes strong influent, keeps the media continuously wet, improves distribution and flushes excess biofilm.
- **Depth** — rock 1–2.5 m (weight-limited); plastic towers 6–12 m.
- **Temperature effect** — performance falls in cold weather because biological rates and natural ventilation both drop.

Worked example: $Q = 2,000 \text{ m}^3\text{/d}$ onto a filter 20 m in diameter ($A = 314 \text{ m}^2$) $\rightarrow$ hydraulic loading = $6.4 \text{ m}^3\text{/m}^2\cdot\text{d}$ (intermediate-rate).

● Classification by Loading Rate

Trickling filters are classified by how hard they are loaded — this sets performance.

Type	Hydraulic loading (m ³ /m ² ·d)	Organic loading (kg BOD/m ³ ·d)	BOD removal	Recirculation
Low-rate (standard)	1 – 4	0.08 – 0.32	80 – 90%	Usually none
Intermediate-rate	4 – 10	0.24 – 0.48	50 – 70%	Sometimes
High-rate	10 – 40	0.32 – 1.0	65 – 85%	Always
Super-rate (plastic tower)	12 – 70	0.8 – 6.0	65 – 85%	Usually
Roughing filter	40 – 200	1.5 – 6.0+	40 – 65%	Not needed

Note the roughing filter: deliberately under-treats — it is used ahead of another process to knock the peak load off a strong industrial effluent.

● Trickling Filter — Operating Problems

Symptom → cause → cure, in the same diagnostic style as Part 2.

Problem	Cause	Corrective action
Ponding (water standing on top)	Excess biofilm or debris clogging voids; media too small	Increase hydraulic flushing, raise recirculation, rake/dose surface; ensure good primary screening
Odour	Anaerobic conditions from overloading or poor ventilation	Reduce organic load, restore ventilation, keep underdrains free
Filter flies (Psychoda)	Dry spots and thin film at the edges	Continuous even wetting, higher recirculation, occasional flooding
Poor BOD removal	Overloading, short-circuiting, uneven distribution	Check distributor arms and orifices; reduce loading
Icing (cold climates)	Low temperature, exposure	Cover the filter, reduce recirculation in winter
Sloughing surge	Seasonal or sudden mass detachment	Ensure clarifier has capacity to handle the humus load

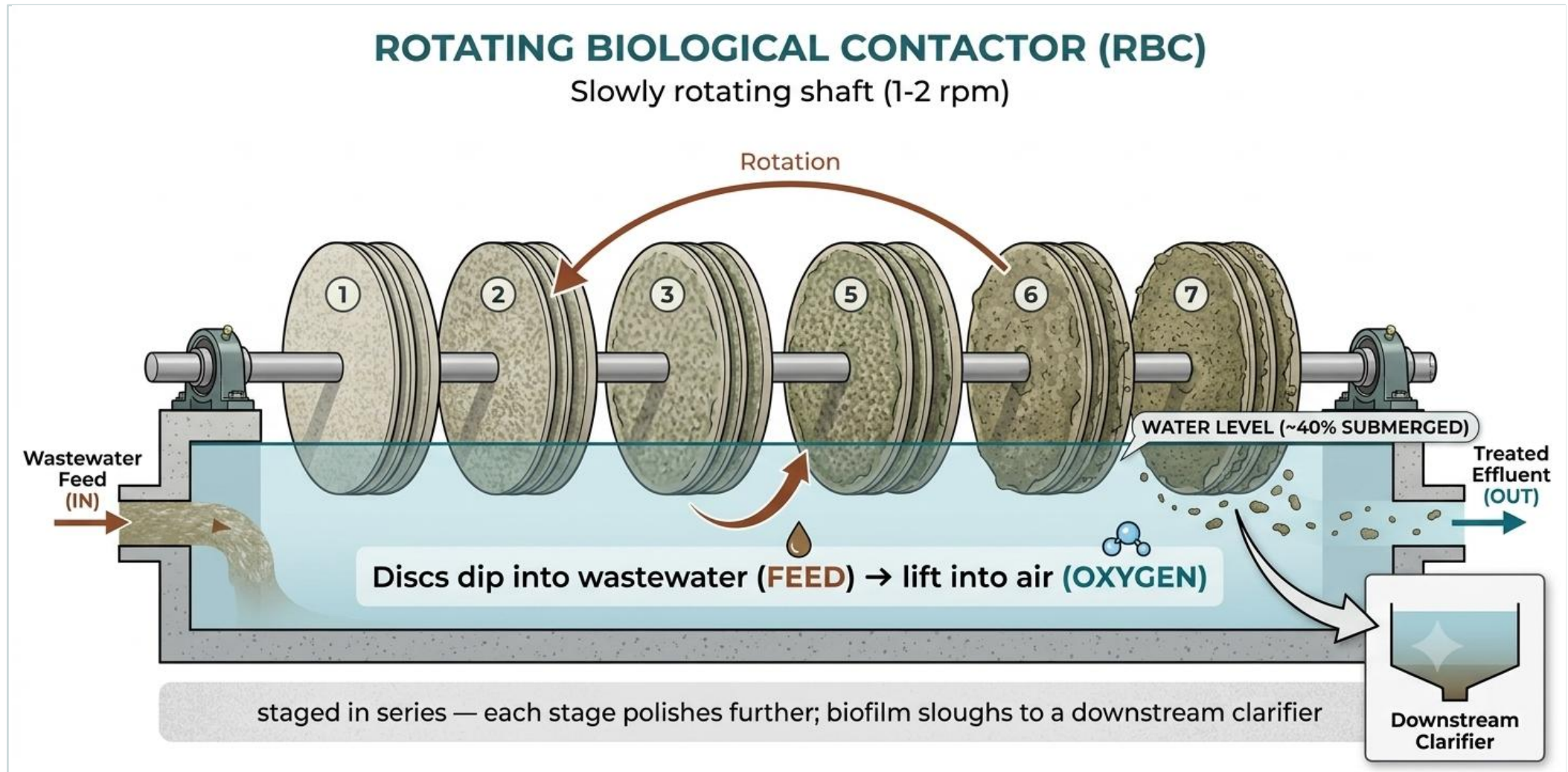
PART C

Rotating Biological Contactors

Discs · staging · design loading

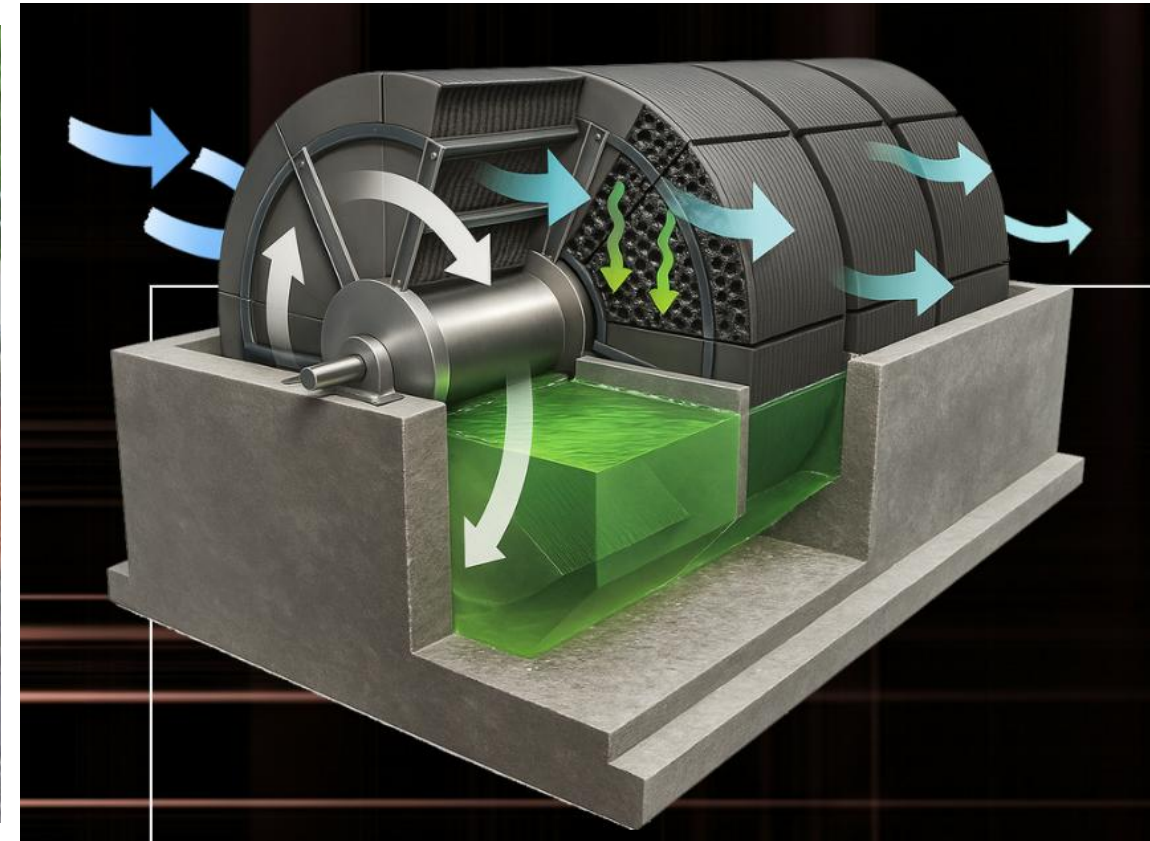
● The Rotating Biological Contactor (RBC)

Biofilm on slowly rotating discs — alternately fed and aerated by the rotation itself.



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Biofilm on slowly rotating discs — alternately fed and aerated by the rotation itself.



● RBC — Construction and Operation

An elegantly simple idea: move the biofilm between the water and the air.

- **Construction** — large-diameter (2–3.7 m) closely-spaced plastic discs mounted on a horizontal shaft in a contoured tank; about 40% of each disc is submerged.
- **Rotation** — the shaft turns slowly (1–2 rpm, ~ 0.3 m/s peripheral speed), driven mechanically or by air.
- **The cycle** — submerged half absorbs BOD from the wastewater; the exposed half takes up oxygen from the air. Each rotation repeats the feed–aerate cycle.
- **Staging** — units are arranged in 3–4 stages in series; the first stage carries the heaviest load, later stages polish and can nitrify.
- **Shear controls the film** — rotation shears off excess biofilm, which is carried to a downstream clarifier.
- **Design loading** — about $4\text{--}10$ g BOD/m²·d of disc area (lower, 2–5, if nitrification is required); hydraulic loading $0.03\text{--}0.08$ m³/m²·d.

● RBC — Strengths and Weaknesses

Popular for small and medium flows; less common at large scale.

Advantages	Limitations
Very low energy — no blowers, only a slow drive motor	Mechanical: shaft and bearing failures under heavy biofilm load
Simple to operate; little skilled attention needed	Sensitive to overloading, especially the first stage
Compact footprint compared with a trickling filter	Higher capital cost per unit capacity
Short start-up; recovers quickly after upsets	Media/disc replacement is expensive
Staging gives progressively cleaner effluent and nitrification	Usually needs covering (weather, algae, odour)
No sludge return, no bulking problems	Performance drops in cold weather

Practical note: load the FIRST stage carefully — most RBC failures begin with an overloaded, oxygen-starved lead stage.

PART D

Modern Hybrid Systems

MBBR · IFAS · packed-bed reactors

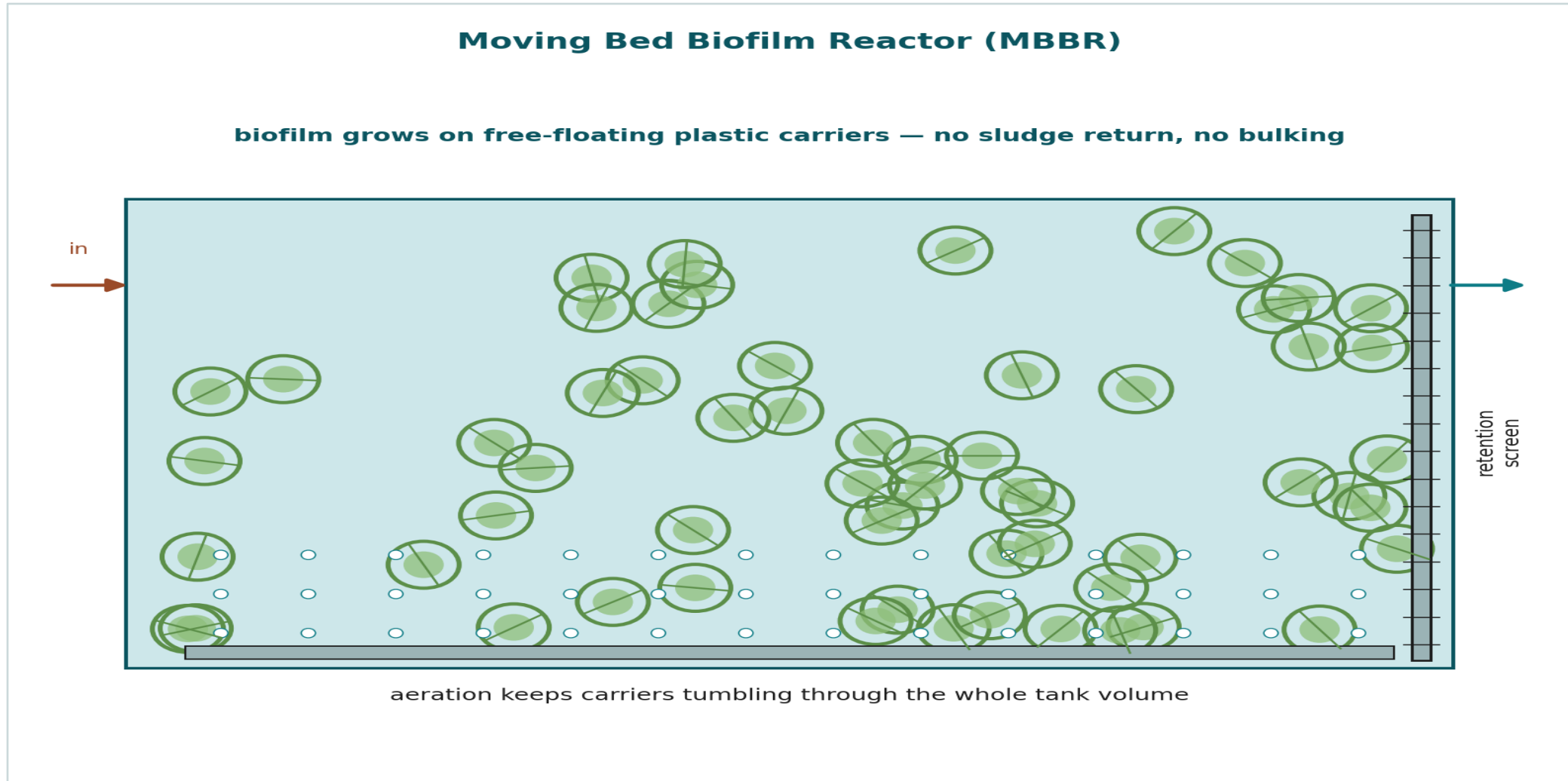
● Moving Bed Biofilm Reactor (MBBR)

The modern answer: keep the biofilm, but let the media move through the whole tank.



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The modern answer: keep the biofilm, but let the media move through the whole tank.



● MBBR — How It Works and Why It Spread

Developed in Norway in the late 1980s; now one of the most widely used industrial processes.

- **The carriers** — small plastic elements (typically 10–25 mm) with a large protected internal surface, 350–1,200 m²/m³, filled to 30–70% of the tank volume.
- **Kept moving** — aeration (or a mixer in anoxic zones) keeps the carriers tumbling, so the biofilm contacts the wastewater throughout the whole tank.
- **Retention screens** — sieves at the outlet keep the carriers in the reactor while the water passes.
- **No RAS, no bulking** — biomass stays on the carriers, so there is no sludge return line and settleability problems disappear as a control concern.
- **Design basis** — loading is expressed per unit of carrier surface area (g BOD/m²·d), not per tank volume — typically 5–15 g BOD/m²·d for carbon removal, 1–3 for nitrification.
- **Still needs solids separation** — sloughed biofilm is removed downstream by a clarifier, DAF or filter.

● IFAS — Upgrading an Existing Plant

Integrated Fixed-film Activated Sludge: put carriers INTO an activated-sludge tank.

- **The concept** — carriers are added to an existing aeration tank, so suspended flocs AND attached biofilm work side by side in the same volume.
- **Why it is powerful** — total biomass rises sharply without enlarging the tank, so a plant that is hydraulically or organically overloaded can be upgraded with minimal civil works.
- **Nitrification retrofit** — slow nitrifiers live safely on the carriers regardless of the suspended sludge age, so an existing plant can be made to nitrify without extending SRT.
- **What is retained** — the clarifier and RAS line stay, because the suspended fraction still has to settle and return.
- **Typical result** — capacity increases of 50–100% within the same tank footprint.

The industrial appeal: an overloaded tannery activated-sludge plant can often be rescued by adding carriers instead of building new tanks.

- IFAS — Upgrading an Existing Plant



● Comparing the Attached-Growth Processes

Selection is a trade-off between footprint, energy, robustness and cost.

Process	Energy	Footprint	Operator skill	Best suited to
Trickling filter (rock)	Very low (natural draught)	Large	Low	Small towns, low-cost schemes
Plastic-media tower	Low	Moderate	Low	Roughing strong industrial effluent
RBC	Low	Compact	Low	Small–medium flows, staged nitrification
MBBR	Moderate (aeration)	Compact	Moderate	Industrial, variable and toxic loads
IFAS	Moderate	Uses existing tank	Moderate	Upgrading an overloaded ASP

The pattern: classical units (TF, RBC) minimise energy and skill but need land; modern carrier systems minimise land and maximise robustness at some energy cost.

● Attached versus Suspended Growth — the Verdict

Bringing Parts 2 and 3 together.

Feature	Suspended growth (ASP)	Attached growth (biofilm)
Where biomass lives	Free-floating flocs in mixed liquor	Fixed film on media or carriers
Biomass retention	By clarifier + RAS return	Inherent — cannot wash out
Sludge age control	Set by operator (wasting rate)	Self-regulating by sloughing
Bulking risk	Yes — the main failure mode	Essentially none
Shock / toxic tolerance	Lower — biomass fully exposed	Higher — film is protected, deeper layers survive
Operator skill needed	High	Low to moderate
Flexibility of control	High (many dials)	Limited
Footprint	Compact	Larger for classical units; compact for MBBR

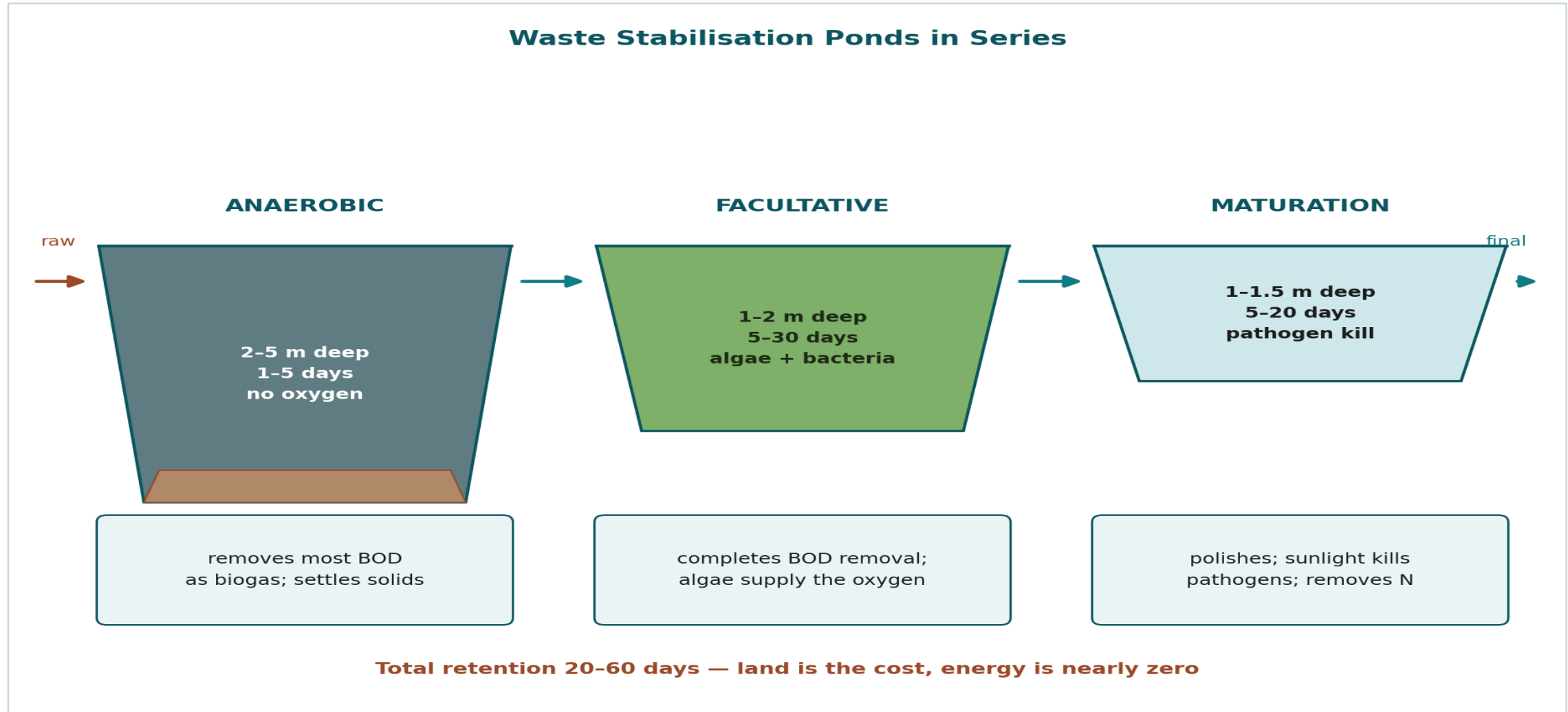
PART A

Lagoons and Stabilisation Ponds

Anaerobic · facultative · maturation · aerated lagoons

● Waste Stabilisation Ponds — the Standard Series

Three pond types in sequence, each doing a different job.



● The Three Pond Types

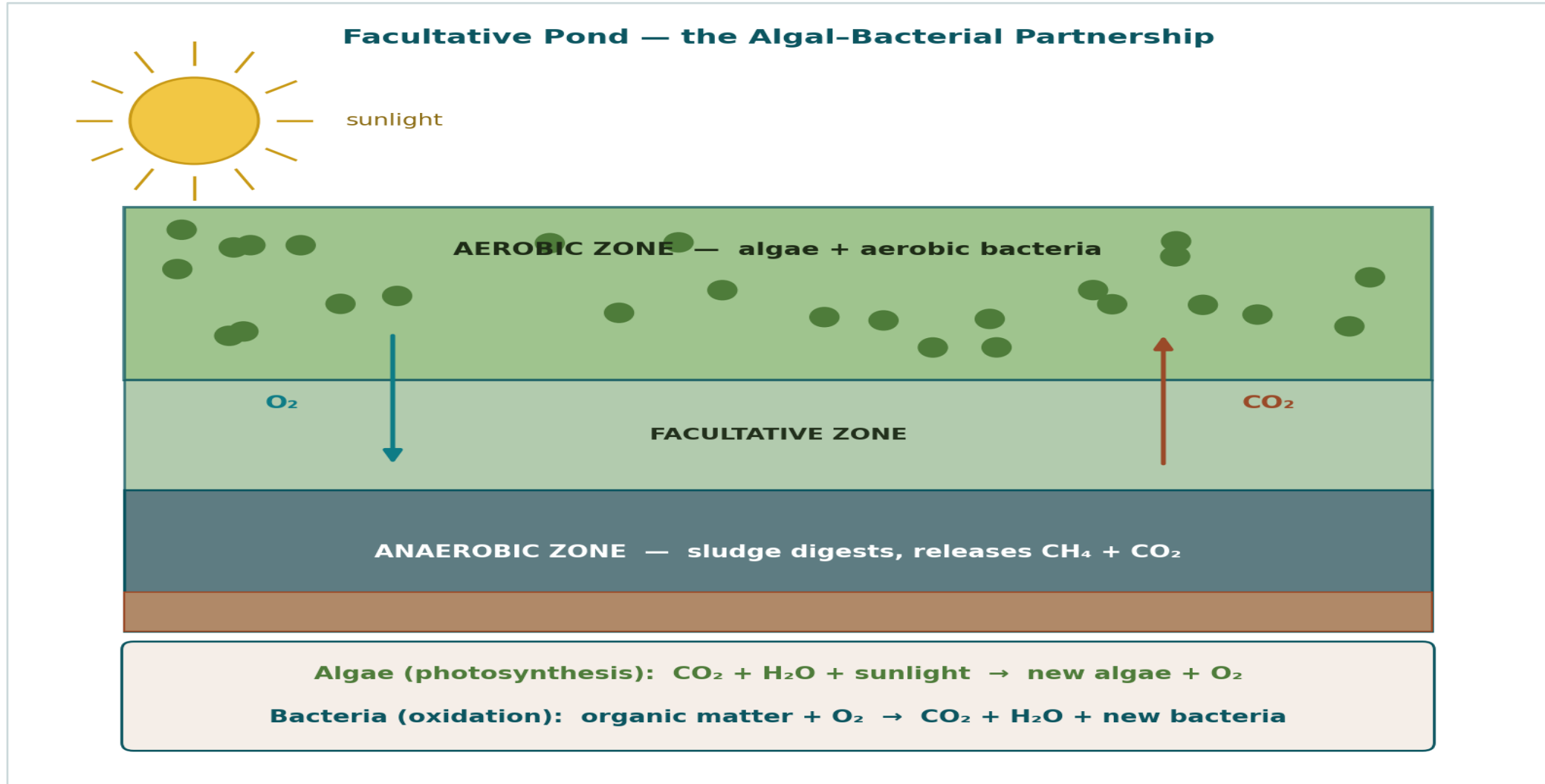
What happens in each stage of the series.

- **Anaerobic pond** — deep (2–5 m) and heavily loaded so that oxygen is entirely absent. Solids settle and digest anaerobically, releasing methane. Removes 50–70% of BOD in only 1–5 days. Small area, but odour risk if overloaded or if sulphate is high.
- **Facultative pond** — shallow (1–2 m) with an aerobic upper layer and an anaerobic bottom. Algae and bacteria work in partnership (next slide). Retention 5–30 days; the main BOD-removal stage.
- **Maturation pond** — shallow (1–1.5 m), lightly loaded and fully aerobic. Its job is polishing: sunlight and high pH destroy pathogens, and nutrients are removed. Retention 5–20 days.
- **Total retention** — typically 20–60 days across the series, which is why land is the binding constraint.

Design logic: load the ponds hard where it is cheap (anaerobic), then finish gently where quality matters (maturation).

● The Facultative Pond — Algae and Bacteria in Partnership

The elegant mechanism at the heart of pond treatment: sunlight supplies the oxygen.



● Aerated Lagoons

A pond fitted with aerators — the bridge between ponds and activated sludge.

- **What it is** — a basin 3–5 m deep with surface or diffused aerators supplying oxygen mechanically instead of relying on algae.
- **Why** — oxygen supply no longer limits the loading, so far more BOD can be treated in far less land than a facultative pond.
- **Fully-mixed (aerobic) lagoon** — enough power input to keep all solids in suspension; behaves like activated sludge without sludge return, so retention is 3–10 days.
- **Facultative (partially-mixed) lagoon** — less power; solids settle and digest anaerobically on the floor while the upper layer stays aerobic. Cheaper to run.
- **Followed by** — a settling pond or clarifier, because the effluent carries suspended biomass.
- **Trade-off** — needs power and maintenance, but cuts land requirement dramatically compared with stabilisation ponds.

● Aerated Lagoons

A pond fitted with aerators — the bridge between ponds and activated sludge.



● Aerated Lagoons

A pond fitted with aerators — the bridge between ponds and activated sludge.



● Ponds and Lagoons — Strengths and Limitations

Why they persist despite being the oldest technology in the field.

Advantages	Limitations
Lowest capital and operating cost of any process	Very large land requirement — often the deciding factor
Almost no energy (stabilisation ponds)	Performance varies with season, sunlight and temperature
Extremely simple — minimal skilled operation	Long retention: 20–60 days
Excellent pathogen removal in maturation ponds	Algae in the effluent raise suspended solids and BOD
No daily sludge handling (desludge every 2–10 years)	Odour risk from anaerobic ponds, especially with sulphate
Robust; absorbs shock loads by sheer volume	Little process control once built; poor for toxic industrial loads

The algae paradox: the algae that produce the oxygen also leave with the effluent — so a well-working pond can still fail a suspended-solids limit.

PART B

Anaerobic Treatment

The four-stage chain · reactors · UASB

● Why Treat Anaerobically?

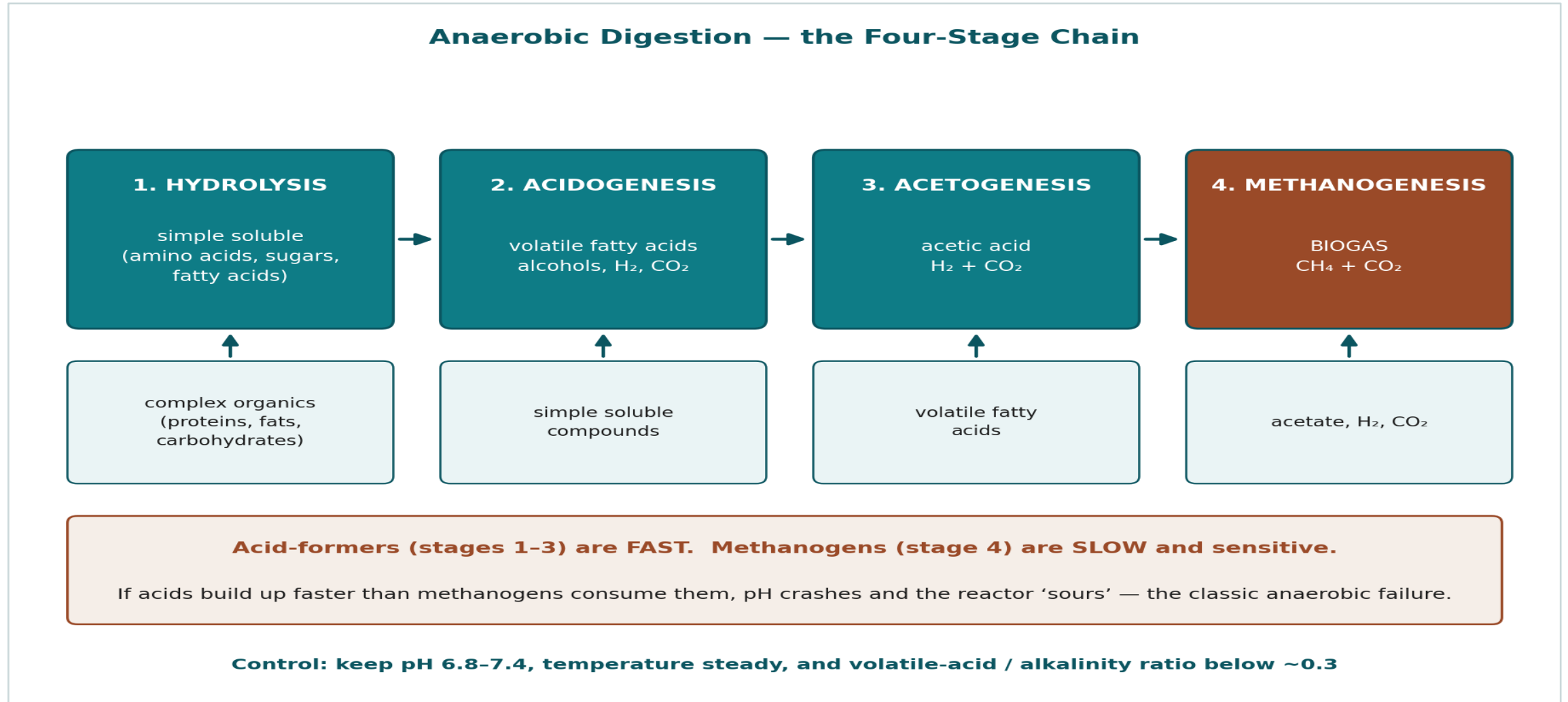
For strong wastewater, removing oxygen from the equation changes the economics entirely.

- **No aeration energy** — aeration is 45–60% of an aerobic plant's energy bill; anaerobic treatment eliminates it completely.
- **It produces energy** — the organic load is converted largely into methane, which can be burned for heat or power.
- **Very little sludge** — the biomass yield is roughly 0.05–0.15 kg VSS per kg COD, compared with 0.4–0.6 for aerobic. Sludge handling costs collapse.
- **Handles high strength** — aerobic systems struggle above about 1,500–2,000 mg/L COD because the oxygen demand becomes impractical; anaerobic systems thrive there.
- **Compact (high-rate reactors)** — modern UASB and EGSB designs achieve very high volumetric loadings.
- **The catch** — it does NOT produce a dischargeable effluent. Anaerobic treatment is a pre-treatment: an aerobic polishing stage almost always follows.

Position it correctly: anaerobic first to strip the bulk load and make gas, aerobic afterwards to meet the discharge limit.

● Anaerobic Digestion — the Four-Stage Chain

Four groups of micro-organisms working in sequence; each depends on the one before.



● The Four Stages in Detail

What each group of organisms does — and why the balance between them is delicate.

- 1 • Hydrolysis** — extracellular enzymes break insoluble polymers (proteins, fats, carbohydrates) into soluble monomers: amino acids, sugars, long-chain fatty acids. For particulate or fatty wastes this is the RATE-LIMITING step.
- 2 • Acidogenesis (fermentation)** — fast-growing acid-forming bacteria ferment the monomers into volatile fatty acids (VFAs), alcohols, hydrogen and CO₂.
- 3 • Acetogenesis** — acetogens convert the longer VFAs into acetic acid, hydrogen and CO₂. This step is inhibited by hydrogen, so it depends on methanogens removing the H₂.
- 4 • Methanogenesis** — slow-growing methanogenic archaea convert acetate (≈70% of methane) and H₂/CO₂ (≈30%) into methane. Doubling time is days, not hours.
- The imbalance risk** — stages 1–3 are fast, stage 4 is slow. If loading rises abruptly, VFAs accumulate, alkalinity is consumed, pH falls, and methanogens are inhibited further — a self-accelerating failure called SOURING.

● Anaerobic Operating Conditions

Methanogens are the fussiest organisms in wastewater treatment — these are their requirements.

Parameter	Optimum / range	Why it matters
pH	6.8 – 7.4	Methanogens inhibited below ~6.5; souring is irreversible if unchecked
Temperature (mesophilic)	30 – 38 °C	The usual operating band; Bangladesh's climate suits it well
Temperature (thermophilic)	50 – 57 °C	Faster and better pathogen kill, but less stable
Alkalinity	2,000 – 4,000 mg/L as CaCO ₃	Buffers the VFAs produced; the key stability reserve
VFA / alkalinity ratio	< 0.3	The best early-warning indicator of overload
Nutrients COD : N : P	≈ 350 : 5 : 1	Far less nitrogen needed than aerobic (100:5:1)
Toxicity limits	Sulphide < 200 mg/L; heavy metals low	Directly relevant to tannery effluent

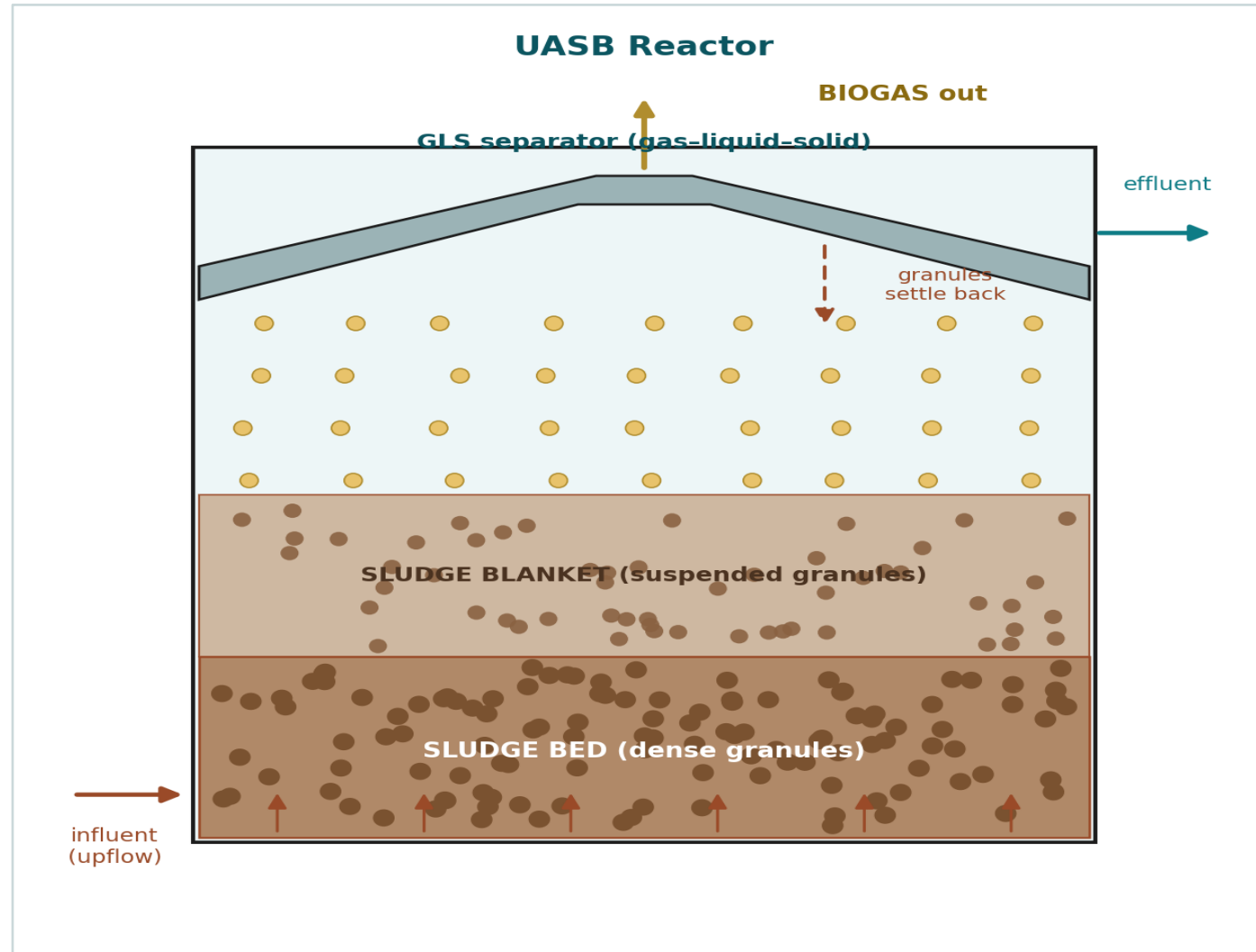
● Anaerobic Reactor Types

The evolution from simple tanks to high-rate reactors that retain biomass.

Reactor	Principle	Loading (kg COD/m ³ ·d)	Notes
Anaerobic pond	Open basin, long retention	0.1 – 0.4	Cheapest; odour risk
Conventional digester	Stirred tank, biomass with liquid	1 – 3	Used for sludge digestion
Anaerobic contact	Digester + settler + recycle	1 – 5	Biomass returned like RAS
Anaerobic filter	Biofilm on fixed media	5 – 15	Can clog with solids
UASB	Granular sludge blanket, upflow	5 – 15 (up to 25)	The industry workhorse
EGSB	Expanded bed, high upflow velocity	15 – 30	For dilute or cool wastewater
AnMBR	Anaerobic + membrane retention	10 – 25	Clear effluent; membrane cost

● The UASB Reactor

Upflow Anaerobic Sludge Blanket — the most widely used high-rate anaerobic reactor.



● UASB — How It Works

Three features make the UASB work: granules, upflow, and the GLS separator.

- **Upflow distribution** — influent is fed evenly across the floor and flows upward through the sludge bed; typical upflow velocity 0.5–1.5 m/h.
- **The granules** — under the right conditions the biomass self-organises into dense granules 1–3 mm across that settle rapidly (30–80 m/h). This is what allows huge biomass to be retained in a small volume.
- **Sludge bed and blanket** — a dense granular bed at the bottom, and above it a looser blanket of granules suspended by rising gas bubbles.
- **The GLS separator** — the gas–liquid–solid device at the top collects biogas in an inverted hood, allows granules to fall back, and lets clarified effluent leave. Without it, granules would wash out.
- **Typical performance** — 70–85% COD removal at loadings of 5–15 kg COD/m³·d with HRT of only 6–12 hours.
- **Start-up is the hard part** — granule formation takes 2–6 months unless seeded with granular sludge from an existing reactor.

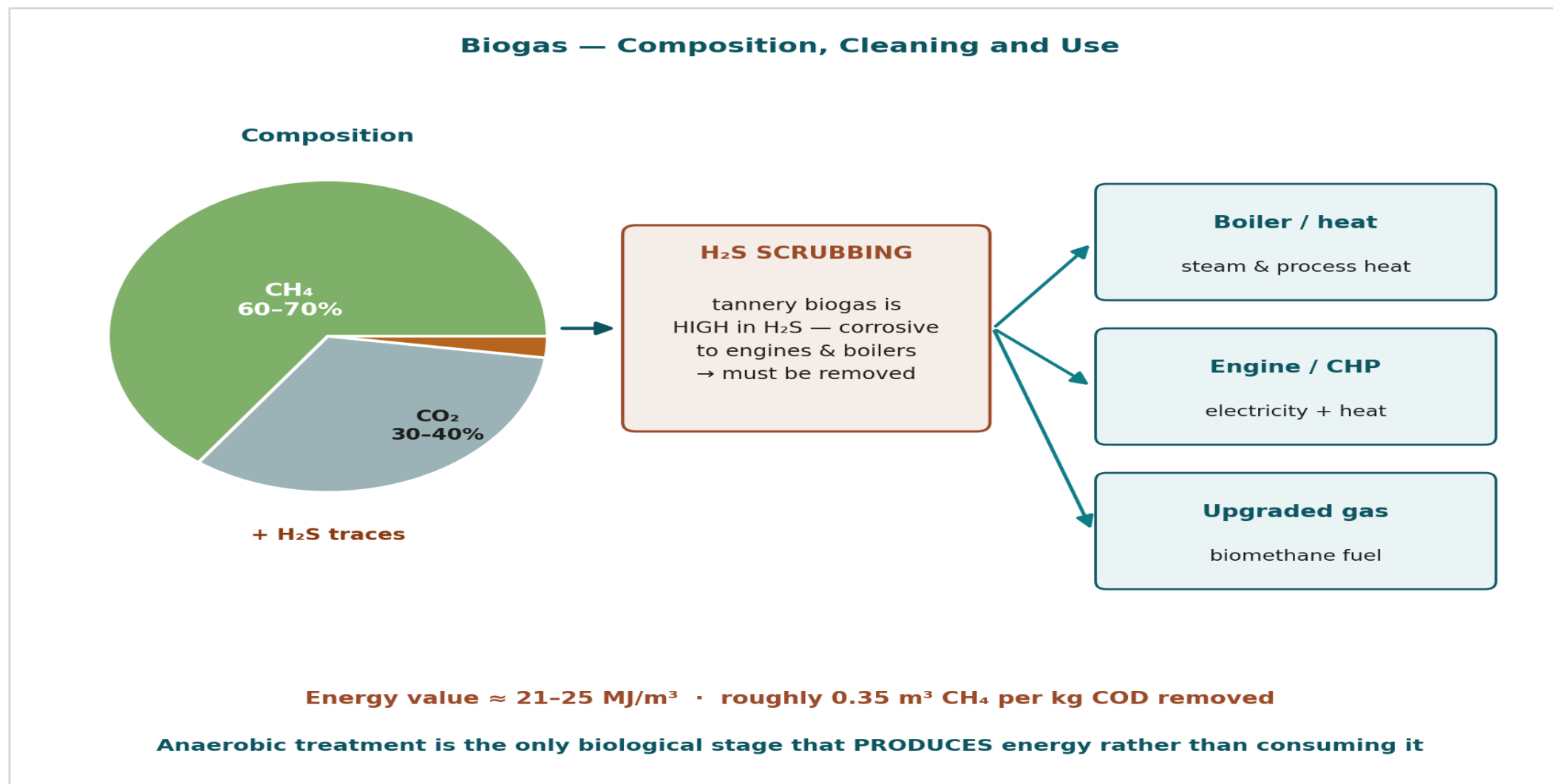
PART C

Biogas and Energy Recovery

Composition · yield · H_2S · utilisation

● Biogas — Composition, Cleaning and Use

The product that turns a treatment cost into an energy asset.



● Biogas — the Numbers

How much gas, how much energy, and what it is worth.

Theoretical yield $\approx 0.35 \text{ m}^3 \text{ CH}_4$ per kg COD removed (at standard conditions)

- **Composition** — 60–70% methane, 30–40% carbon dioxide, plus traces of H_2S , N_2 and water vapour.
- **Calorific value** — roughly 21–25 MJ/ m^3 of raw biogas, about 60% that of natural gas.
- **Practical yield** — somewhat below theoretical, because part of the COD goes to new biomass and some methane dissolves in the effluent.
- **Uses in order of simplicity** — direct combustion in a boiler for process heat; a gas engine or CHP unit for electricity plus heat; or upgrading to biomethane.
- **Flaring** — if gas cannot be used it MUST still be flared, never vented: methane is a far more potent greenhouse gas than CO_2 .

Worked example: 1,000 m^3/d at 4,000 mg/L COD with 75% removal $\rightarrow$ 3,000 kg COD removed $\rightarrow \approx 1,000 \text{ m}^3 \text{ CH}_4/\text{d} \approx 21,000\text{--}25,000$ MJ/d of fuel.

● The H₂S Problem — Critical for Tanneries

Tannery biogas is unusually corrosive, and the reason is sulphate and sulphide.

- **Where it comes from** — tannery effluent is rich in sulphate (from pickling and chrome tanning) and sulphide (from liming). In the reactor, sulphate-reducing bacteria convert sulphate to sulphide, which partitions into the gas as H₂S.
- **Two damages** — H₂S burns to sulphur dioxide and forms sulphuric acid, corroding engines, boilers, gas lines and concrete; and sulphide is directly TOXIC to methanogens above roughly 200 mg/L.
- **Competition for substrate** — sulphate-reducing bacteria consume the same organic matter as methanogens, so a high-sulphate effluent yields LESS methane per kg COD.
- **Removal methods** — iron dosing (precipitates FeS in the reactor), biological scrubbing with a small air dose, activated-carbon or caustic scrubbers, or water scrubbing.
- **Design implication** — sulphide oxidation and stream segregation in PRIMARY treatment directly protect the anaerobic reactor and the value of its gas.

The link back to Part 1: the primary-treatment decisions you make upstream determine whether anaerobic treatment is viable at all.

● Anaerobic versus Aerobic — the Full Comparison

The decision is driven mainly by wastewater strength.

Feature	Anaerobic	Aerobic
Energy	Produces (methane)	Consumes (aeration)
Sludge yield	0.05 – 0.15 kg VSS/kg COD	0.4 – 0.6 kg VSS/kg COD
Nutrient demand (COD:N:P)	≈ 350 : 5 : 1	≈ 100 : 5 : 1
Best influent strength	High (> 1,500–2,000 mg/L COD)	Low to medium
Effluent quality	Partial — needs polishing	Meets discharge standards
Start-up time	2 – 6 months (granulation)	2 – 4 weeks
Temperature sensitivity	High — needs 30–38 °C	Moderate
Odour / gas hazard	H ₂ S, methane — must be managed	Low
Robustness to toxins	Lower (methanogens sensitive)	Higher

● Application to Tannery Effluent

Where these processes fit — and where they do not — in a tannery scheme.

- **Anaerobic pre-treatment fits the strength** — beamhouse and composite tannery effluent carry high COD, exactly the range where anaerobic treatment is economical.
- **But sulphide is the obstacle** — untreated, it poisons the methanogens and contaminates the gas. Sulphide oxidation upstream is not optional.
- **And salinity inhibits** — high TDS suppresses methanogens; the biomass must be acclimatised, and very high salinity may rule anaerobic treatment out.
- **Ponds in Bangladesh practice** — anaerobic and facultative ponds remain common at CETPs and smaller tanneries because of low cost and the warm climate; the constraints are land and odour.
- **The realistic train** — equalisation and sulphide oxidation → (anaerobic reactor or pond, with gas capture) → aerobic polishing → clarification → tertiary.
- **Chrome first** — chromium must be recovered from the tanyard stream before any biological stage; metals inhibit anaerobic biomass strongly.

PART A

Biological Nutrient Removal

Why · nitrification · denitrification · phosphorus

● Why Nutrients Must Be Removed

Removing BOD alone can still leave a discharge that destroys the receiving water.

- **Eutrophication** — nitrogen and phosphorus are fertilisers. Discharged to a river or lake they trigger algal blooms; when the algae die and decompose, the decay strips dissolved oxygen and fish suffocate.
- **Ammonia toxicity** — un-ionised ammonia (NH_3) is directly toxic to fish at well under 1 mg/L, and the effect worsens at high pH.
- **Oxygen demand** — ammonia also exerts a nitrogenous oxygen demand in the receiving water, roughly 4.6 kg O_2 per kg of nitrogen.
- **Where it comes from in tanneries** — deliming with ammonium salts, plus protein breakdown from hair and hide, make tannery effluent unusually high in ammonia nitrogen.
- **Regulatory direction** — nitrogen and phosphorus limits are tightening worldwide, and are increasingly applied to industrial discharges.

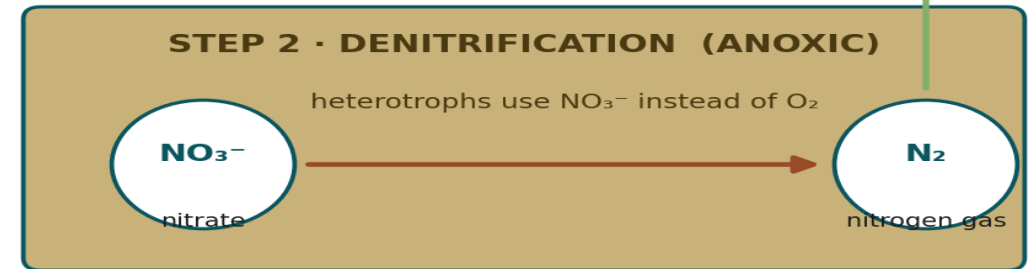
● Nitrogen Removal — the Two-Step Pathway

Nitrogen is removed as a GAS, which is why it leaves neither in the effluent nor in the sludge.

Biological Nitrogen Removal — the Two-Step Pathway



needs O₂ (4.6 kg O₂/kg N) and LONG sludge age (SRT ≥ 10 d)



needs NO oxygen but DOES need carbon (BOD) as electron donor

Overall: $\text{NH}_4^+ \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^- \rightarrow \text{N}_2 \uparrow$

Nitrogen leaves the plant as harmless gas — not in the sludge, not in the effluent.

Key design consequence: the plant needs BOTH an aerobic zone AND an anoxic zone — in space (zones) or in time (SBR phases)

● Step 1 — Nitrification

Slow, oxygen-hungry autotrophs convert ammonia to nitrate.



- **Who does it** — autotrophic bacteria: Nitrosomonas oxidises ammonium to nitrite, Nitrobacter oxidises nitrite to nitrate. They obtain energy from the nitrogen, not from organic matter.
- **They grow slowly** — doubling time is measured in days, not hours. This is the single most important operational fact about nitrification.
- **Requires long sludge age** — SRT must exceed roughly 10 days (more in cold weather), or the nitrifiers are washed out faster than they reproduce.
- **Oxygen demand** — about 4.6 kg O₂ per kg of ammonia nitrogen oxidised: a substantial addition to the aeration load.
- **Consumes alkalinity** — roughly 7.1 kg as CaCO₃ per kg N. If alkalinity is insufficient the pH falls and nitrification stalls.
- **Sensitive** — inhibited by low DO (< 2 mg/L), pH outside 7.0–8.5, low temperature, and by many industrial toxins including sulphide and metals.

● Step 2 — Denitrification

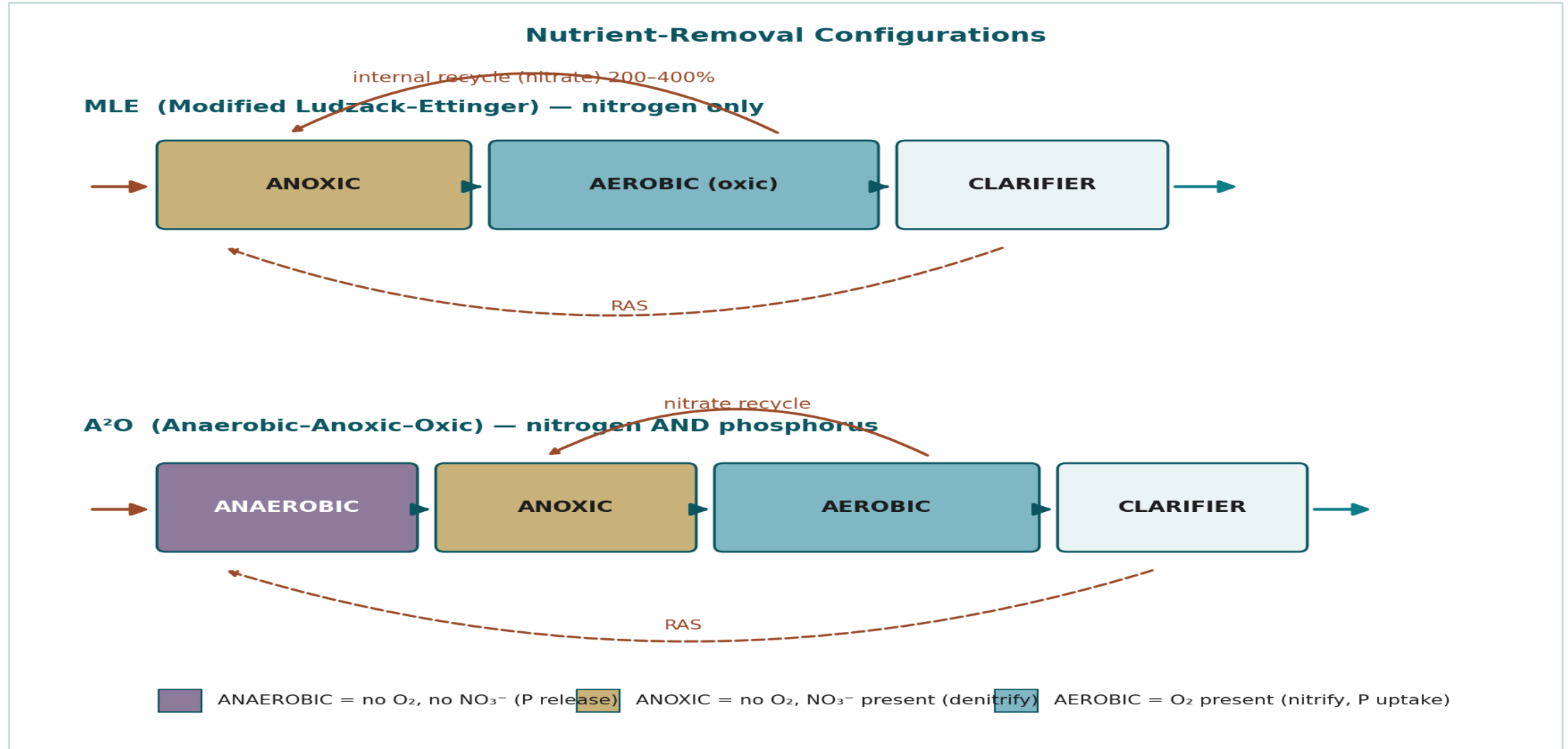
Ordinary heterotrophs breathe nitrate instead of oxygen, releasing nitrogen gas.



- **Who does it** — common heterotrophic bacteria already present in the sludge. They are facultative: given no oxygen but plenty of nitrate, they use nitrate as the electron acceptor.
- **The condition is ANOXIC** — meaning no dissolved oxygen but nitrate present. This is different from anaerobic (neither oxygen nor nitrate).
- **Needs a carbon source** — about 4 kg BOD per kg of nitrate nitrogen reduced. Using the incoming wastewater's own BOD avoids buying methanol.
- **Recovers alkalinity** — returns roughly half the alkalinity that nitrification consumed (about 3.6 kg as CaCO_3 per kg N), helping stabilise pH.
- **Recovers oxygen** — using nitrate as an electron acceptor effectively recovers part of the aeration energy spent on nitrification.
- **Placement matters** — putting the anoxic zone FIRST (pre-denitrification) lets it use the raw influent BOD, which is why MLE is the standard layout.

● Arranging the Zones — MLE and A²O

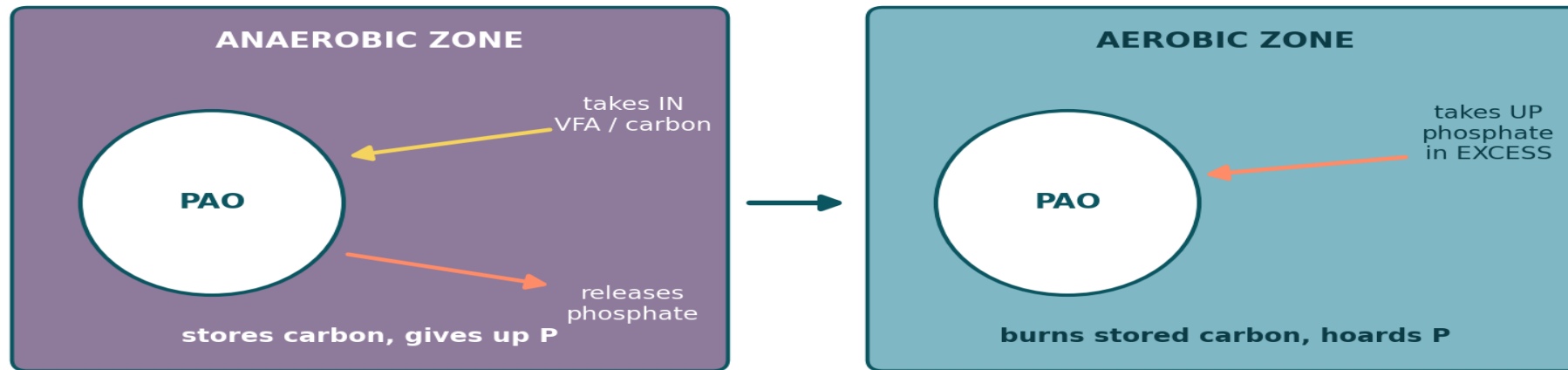
The same biology; what differs is the ORDER of conditions the sludge passes through.



● Biological Phosphorus Removal

Phosphorus cannot become a gas — so it must be removed inside the sludge.

Biological Phosphorus Removal — How PAOs Are Tricked



‘Luxury uptake’: the PAO absorbs far more phosphorus than it needs to grow

The phosphorus is now INSIDE the cells — so wasting the sludge removes it from the water.

No chemicals needed — but the anaerobic zone must be genuinely free of oxygen AND nitrate.

● Phosphorus Removal — Biological or Chemical

Two routes to the same end: get the phosphorus out of the water.

- **Enhanced biological P removal (EBPR)** — PAOs are cycled through anaerobic then aerobic conditions; they release phosphate while storing carbon anaerobically, then take up phosphate in excess ('luxury uptake') aerobically. Wasting the sludge removes the P.
- **The critical requirement** — the anaerobic zone must contain neither oxygen NOR nitrate. Nitrate returning in the RAS is the most common cause of EBPR failure.
- **Needs readily-biodegradable carbon** — volatile fatty acids in the influent feed the PAOs; a weak or already-fermented influent gives poor performance.
- **Chemical P removal** — dose alum, ferric chloride or lime to precipitate phosphate. Simple and reliable, but adds chemical cost and increases sludge volume.
- **Practical choice** — many plants run EBPR as the primary route with chemical dosing as backup polishing to meet a tight limit.

Remember the asymmetry: nitrogen leaves as a gas; phosphorus leaves only in the wasted sludge. So P removal always means more sludge to handle.

PART B

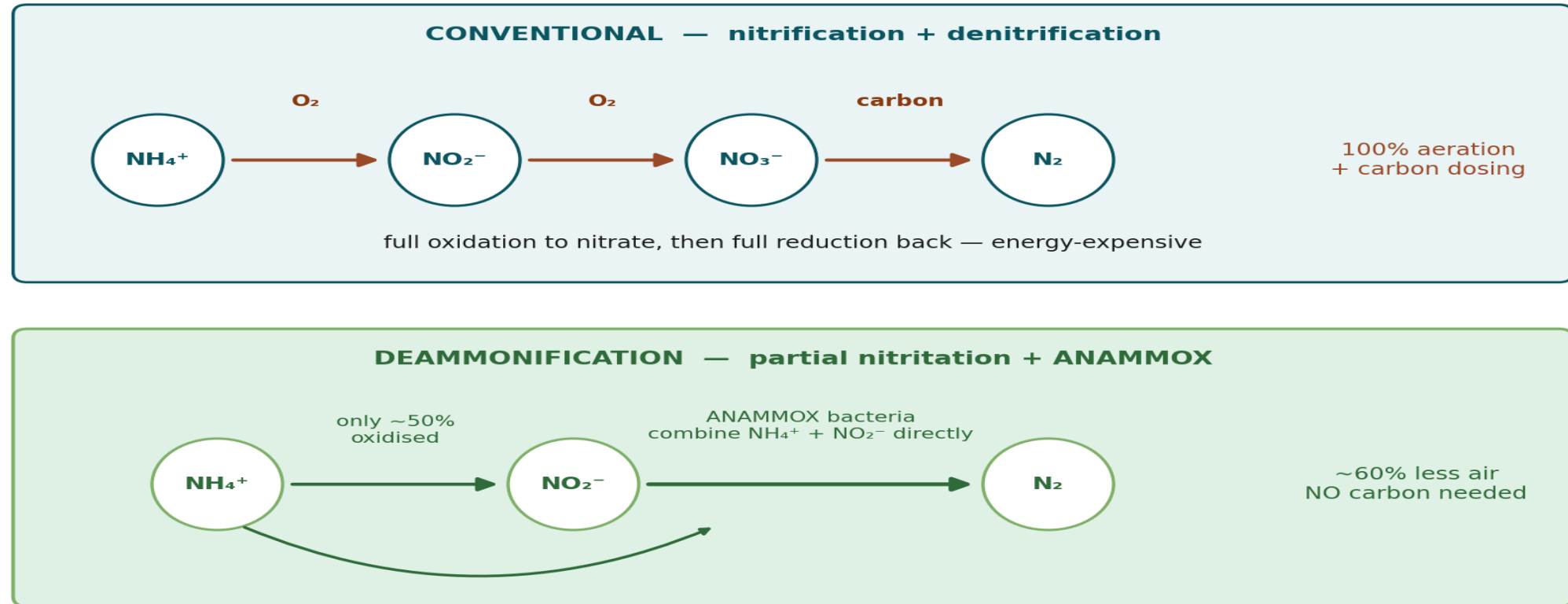
Advanced & Emerging Processes

ANAMMOX · MBR · aerobic granular sludge · bioaugmentation

● ANAMMOX — the Nitrogen Shortcut

A different route to nitrogen gas that skips most of the aeration and all of the carbon.

Conventional N-Removal versus Deammonification (ANAMMOX)



Trade-off: major energy saving, but ANAMMOX bacteria grow very slowly and need warm, stable conditions

● ANAMMOX / Deammonification — Detail

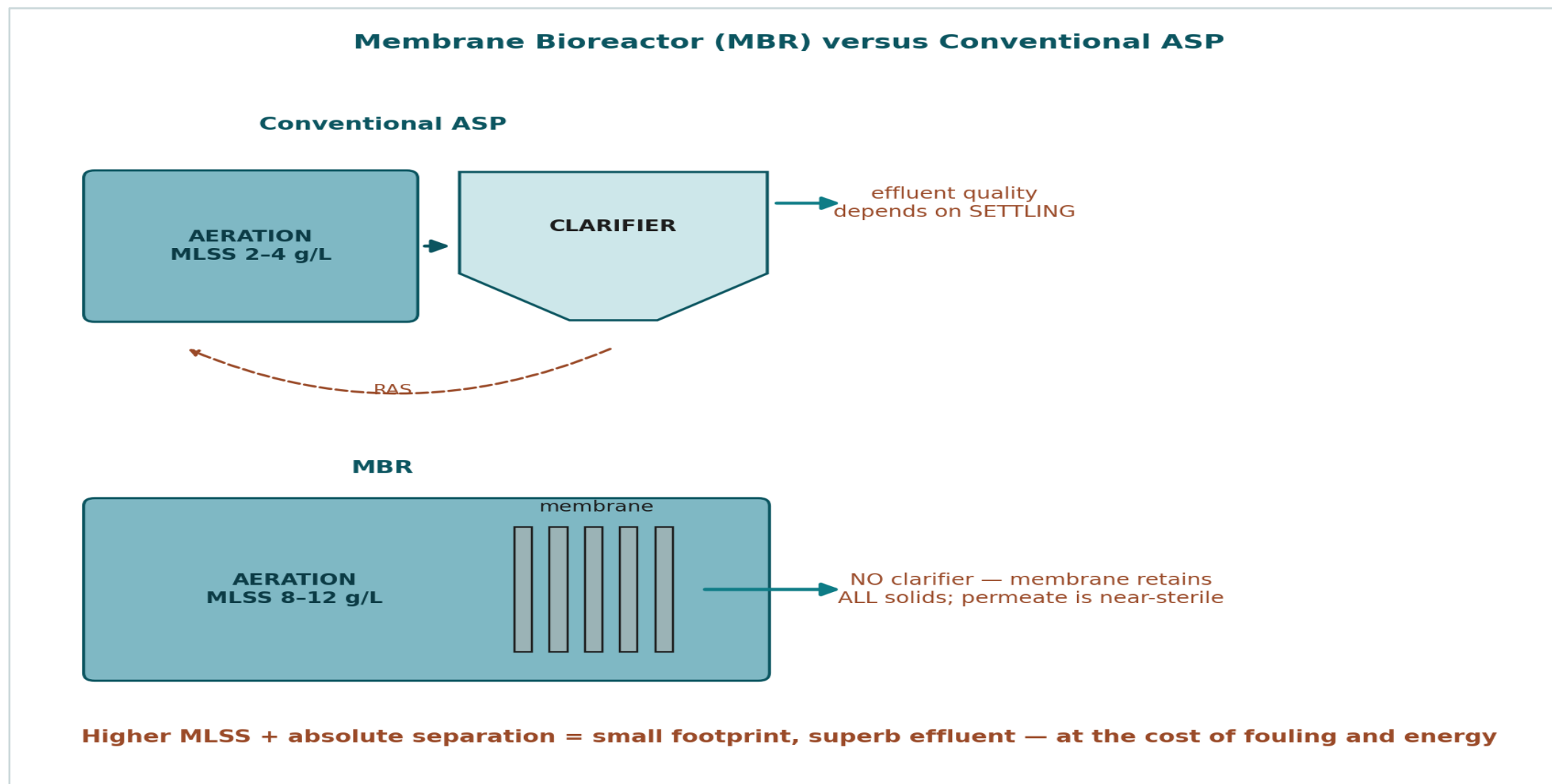
Discovered in the 1990s; now the standard for treating concentrated ammonia streams.



- **The organisms** — anaerobic ammonium-oxidising bacteria (Planctomycetes) combine ammonium and nitrite directly into nitrogen gas, without oxygen and without organic carbon.
- **Deammonification in practice** — partial nitrification first oxidises only about half the ammonium to nitrite, then ANAMMOX combines the two halves.
- **The savings** — roughly 60% less aeration energy, no external carbon dosing, and about 90% less excess sludge than conventional nitrification/denitrification.
- **The difficulty** — ANAMMOX bacteria have doubling times of 10–14 days, so start-up takes months and washout is easy; they need stable temperature (typically 30–35 °C) and careful control of nitrite, which is itself inhibitory.
- **Where it is used** — sidestream treatment of digester liquors, landfill leachate, and other warm, concentrated, low-carbon ammonia streams.
- **Tannery potential** — attractive for concentrated deliming liquors; the warm climate helps, but sulphide and salinity inhibition must be managed.

● Membrane Bioreactor (MBR)

Replace gravity settling with absolute physical separation — and the constraints change completely.



● MBR — Why It Changes the Design Rules

Removing the clarifier removes the constraint that has shaped activated sludge since 1914.

- **The principle** — micro- or ultrafiltration membranes (0.04–0.4 μm) submerged in the aeration tank, or in a side-stream loop, separate treated water from biomass by filtration rather than settling.
- **Settleability becomes irrelevant** — SVI, bulking and sludge blanket control simply cease to be design constraints. This is the fundamental change.
- **Very high MLSS** — 8,000–12,000 mg/L instead of 2,000–4,000, so the same treatment happens in roughly a third of the tank volume.
- **Long SRT, small tank** — SRT and HRT are fully decoupled, so nitrification is easy to sustain even in a compact reactor.
- **Effluent quality** — essentially free of suspended solids and bacteria; ready to feed reverse osmosis for water reuse.
- **The costs** — membrane fouling requiring air scouring and periodic chemical cleaning; higher energy; membrane replacement every 5–10 years; skilled operation.

● Aerobic Granular Sludge

Make the sludge settle in minutes instead of hours — and the clarifier disappears.

- **What it is** — activated sludge cultivated so that the biomass self-assembles into dense, compact granules (0.5–2 mm) rather than loose flocs.
- **How granules are selected** — the reactor is run in batch cycles with a very short settling window; only fast-settling granules are retained, while light flocs are decanted away.
- **Layered structure** — each granule is aerobic at the surface, anoxic beneath, and anaerobic at the core — so carbon, nitrogen AND phosphorus removal occur simultaneously inside a single granule.
- **Performance** — settling velocities of 30–70 m/h and SVI values around 40–60 mL/g; up to ~75% less footprint and ~30% less energy than conventional ASP.
- **No clarifier, no RAS** — settle and decant occur in the same tank, as in an SBR.
- **Status** — commercially proven (the Nereda process); the main challenge is establishing and maintaining stable granulation.

Why it matters: granular sludge attacks the oldest weakness of activated sludge — that separation by gravity is slow and unreliable.

● Bioaugmentation and Salt-Tolerant Biomass

Sometimes the answer is not a new reactor but different organisms.

- **The salinity problem** — high TDS creates osmotic stress that dehydrates and can lyse ordinary microbes; above roughly 10,000–15,000 mg/L TDS, conventional activated sludge deteriorates badly.
- **Acclimatisation** — raising the salinity gradually over several weeks lets the community adapt and select salt-tolerant species. Sudden salinity SHOCKS are far more damaging than steady high salinity.
- **Halotolerant and halophilic cultures** — specialised salt-adapted organisms that continue to degrade organics at TDS levels that inhibit normal sludge.
- **Bioaugmentation** — dosing selected or cultivated strains to accelerate start-up, restore a shocked plant, or degrade specific recalcitrant compounds.
- **Attached growth helps** — biofilm systems (Part 3) hold biomass physically, so salt-stressed organisms cannot be washed out — a practical advantage in saline effluent.

Operational rule: steady conditions beat clever additives — equalisation that prevents salinity swings protects the biology more than any dosed culture.

● Advanced Processes — Comparison

What each one is actually for.

Process	Solves what problem	Main advantage	Main limitation
MLE / A ² O	N (and P) removal	Proven, simple zone layout	Larger volume, recycle pumping
SBR	N and P in one tank	Flexible timing, no clarifier	Automation required
ANAMMOX	High-ammonia, low-carbon streams	~60% less aeration, no carbon	Very slow growth; hard start-up
MBR	Effluent quality & footprint	Superb effluent, reuse-ready	Fouling, energy, cost
MBBR / IFAS	Capacity & robustness	Easy retrofit, shock-tolerant	Carrier screens, moderate energy
Granular sludge	Footprint & settling	Compact, simultaneous C/N/P	Granulation know-how
Bioaugmentation	Toxicity & salinity	No new civil works	Ongoing dosing cost

PART C

Putting It Together — Tannery Design

Constraints · the full train · troubleshooting · the future

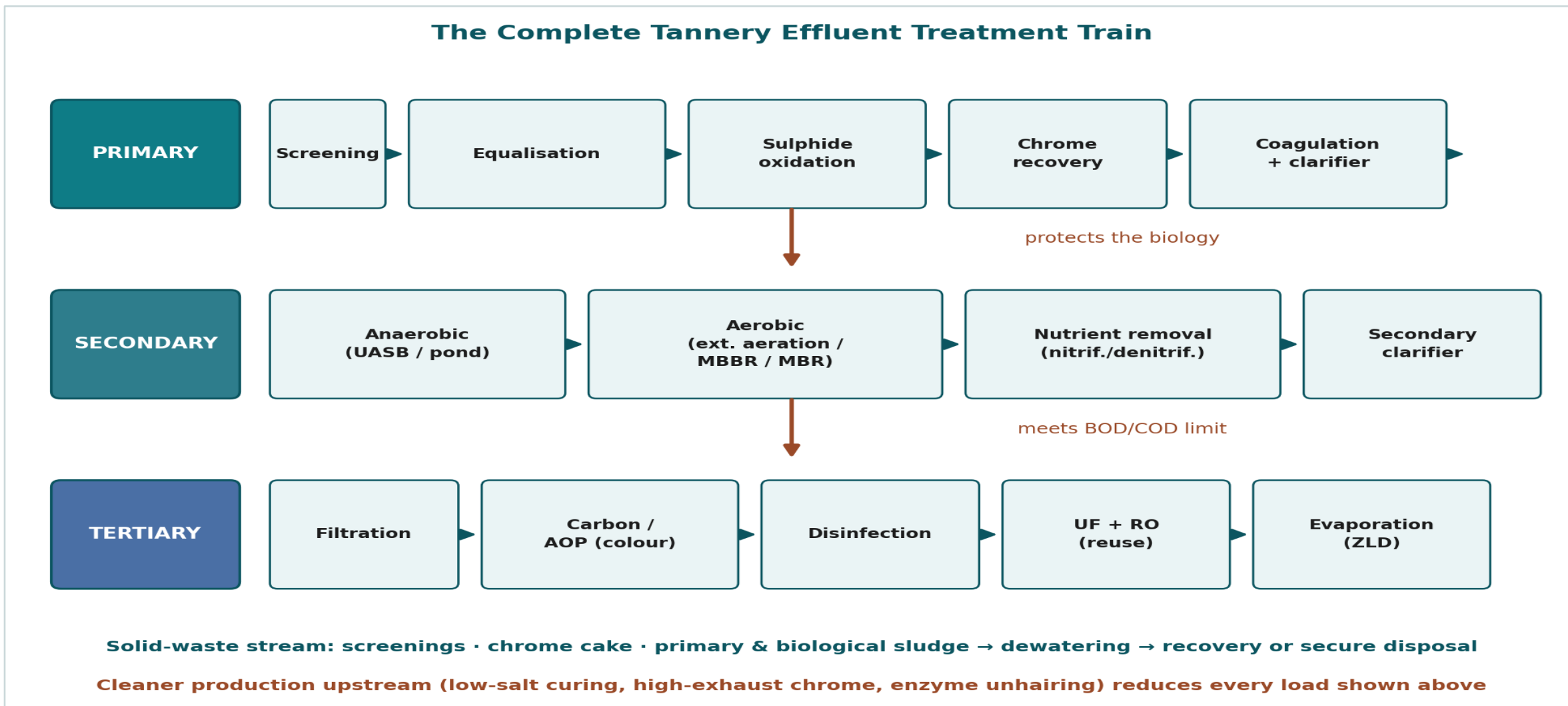
● The Four Constraints of Tannery Effluent

Everything about tannery biological design follows from these.

Constraint	Why it hurts the biology	Where it is solved
High salinity (TDS)	Osmotic stress bursts cells; biological treatment cannot remove salt at all	Acclimatised/halotolerant biomass; salt reduction at source; RO or ZLD in tertiary
Sulphide	Toxic to aerobes AND methanogens; exerts oxygen demand; corrodes; forms H ₂ S gas	Catalytic oxidation in PRIMARY, before mixing
Chromium	Heavy-metal inhibition of biomass; contaminates sludge as hazardous waste	Segregation and chrome recovery in PRIMARY
High & variable load	Shock loads overwhelm or poison the biomass	Strong equalisation; robust, forgiving process choice

● The Complete Tannery Treatment Train

Every unit from Parts 1–5, assembled in order.



● Choosing the Secondary Process for a Tannery

A practical decision sequence, not a formula.

- **1 · How strong is the effluent?** — above roughly 2,000 mg/L COD, put an anaerobic stage (UASB or anaerobic pond) first to strip the load and recover biogas.
- **2 · How much land is available?** — abundant land and low budget favour ponds and lagoons; constrained land favours extended aeration, MBBR or MBR.
- **3 · How skilled is the operating team?** — limited skilled staff strongly favours attached growth or extended aeration over closely-controlled activated sludge.
- **4 · How variable and toxic is the influent?** — frequent shocks favour biofilm systems (MBBR/IFAS), which recover far faster than suspended sludge.
- **5 · Is ammonia limited? Is reuse required?** — ammonia limits demand a long SRT and an anoxic zone; water reuse demands MBR or UF followed by RO.
- **The common answer** — equalisation → (anaerobic) → extended-aeration or MBBR aerobic stage → clarifier → tertiary polishing. Forgiving beats clever.

● Common Effluent Treatment Plants (CETPs)

How the Bangladesh tannery cluster is actually treated — and why CETPs struggle.

- **The concept** — many small tanneries cannot each afford a full plant, so effluent is piped to one shared CETP, as at the Savar tannery estate.
- **Economies of scale** — one professionally-run plant with proper equalisation, sludge handling and monitoring is far better than fifty poor ones.
- **Why CETPs underperform** — hydraulic overload beyond design capacity; no chrome segregation at source, so chromium arrives mixed; sulphide arriving un-oxidised; and a mixed influent that no single design can suit.
- **Sludge is the hidden problem** — mixed chrome-bearing sludge is hazardous waste, and its handling and secure disposal is often the weakest link.
- **What actually fixes it** — enforcement of pre-treatment AT EACH TANNERY (chrome recovery, sulphide oxidation, screening) before discharge to the common sewer.

Engineering reality: a CETP can only be as good as the pre-treatment of the factories feeding it.

● References

- Metcalf & Eddy — Wastewater Engineering: Treatment and Resource Recovery (the principal reference throughout Parts 1–5).
- Grady, Daigger, Love & Filipe — Biological Wastewater Treatment (kinetics, process theory).
- Jenkins, Richard & Daigger — Manual on the Causes and Control of Activated Sludge Bulking and Foaming.
- Rittmann & McCarty — Environmental Biotechnology: Principles and Applications (biofilm and nutrient-removal theory).
- Mara, D. — Domestic Wastewater Treatment in Developing Countries (pond design); van Haandel & Lettinga — Anaerobic Sewage Treatment for Warm Climates.
- van Loosdrecht et al. — aerobic granular sludge and ANAMMOX literature; Judd — The MBR Book.
- Scholz — Key Technologies for Tannery Wastewater Treatment (IULTCS); UNIDO — Treatment of Tannery Effluents; Bangladesh DoE — ECR 2023.